



Virginia State Standards Alignment

For Grades K-8

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About Zearn

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Zearn Math is aligned to Virginia's Mathematics Standards of Learning.

Zearn is committed to ensuring that our high-quality instructional materials align to Virginia's Mathematics Standards of Learning to support Virginia's students and teachers. The tables on the following pages, organized by Virginia Standards of Learning, identify the alignment between each grade-level standard and Zearn Math lesson(s).

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Kindergarten

VIRGINIA'S MATHEMATICS STANDARDS OF LEARNING		ZEARN MATH
Kindergarten Standards		Lessons
Number and Number Sense		
K.NS.1a	Use one-to-one correspondence to determine how many are in a given set containing 30 or fewer concrete objects (e.g., cubes, pennies, balls), and describe the last number named as the total number of objects counted.	Mission 1, Topic B, Lesson 5-6 Mission 1, Topic C, Lesson 7-11 Mission 1, Topic D, Lesson 14-16 Mission 1, Topic E, Lesson 17-22 Mission 1, Topic F, Lesson 23-28 Mission 1, Topic G, Lesson 29 Mission 1, Topic H, Lesson 33-27 Mission 5, Topic A, Lesson 1-5 Mission 5, Topic B, Lesson 6-9 Mission 5, Topic C, Lesson 10-14
K.NS.1b	Recognize and explain that the number of objects remains the same regardless of the arrangement or the order in which the objects are counted.	Mission 1, Topic C, Lesson 7-8, 10 Mission 1, Topic D, Lesson 16 Mission 1, Topic E, Lesson 17-22 Mission 1, Topic F, Lesson 23-28 Mission 1, Topic G, Lesson 31 Mission 1, Topic H, Lesson 36-37 Mission 5, Topic C, Lesson 13-14
K.NS.1c	Represent forward counting by ones using a variety of tools, including five-frames, ten-frames, and number paths (a prelude to number lines).	Mission 1, Topic C, Lesson 7-10 Mission 1, Topic E, Lesson 17, 19, 21 Mission 1, Topic F, Lesson 23, 25-27 Mission 1, Topic G, Lesson 29 Mission 5, Topic A, Lesson 2-5 Mission 5, Topic B, Lesson 6-9 Mission 5, Topic C, Lesson 14 Mission 5, Topic D, Lesson 15-16, 18 Mission 5, Topic E, Lesson 21
K.NS.1d	Count forward orally by ones from 0 to 100.	Mission 5, Topic C, Lesson 13 Mission 5, Topic D, Lesson 15-18
K.NS.1e	Count forward orally by ones, within 100, starting at any given number.	Mission 5, Topic C, Lesson 13 Mission 5, Topic D, Lesson 16-18

VIRGINIA'S MATHEMATICS STANDARDS OF LEARNING		ZEARN MATH
Kindergarten Standards		Lessons
K.NS.1f	Count backward orally by ones when given any number between 1 and 20.	Mission 5, Topic C, Lesson 12-13
K.NS.1g	State the number after, without counting, when given any number between 0 and 30.	Mission 1, Topic D, Lesson 12, 14 Mission 1, Topic E, Lesson 17-22 Mission 1, Topic G, Lesson 29-32 Mission 1, Topic H, Lesson 33-37 Mission 4, Topic H, Lesson 38
K.NS.1h	State the number before, without counting, when given any number between 1 and 20.	Mission 1, Topic D, Lesson 12 Mission 1, Topic H, Lesson 33-36
K.NS.1i	Use objects, drawings, words, or numbers to compose and decompose numbers 11-19 into a ten and some ones.	Mission 5, Topic A, Lesson 2-5 Mission 5, Topic B, Lesson 6-9 Mission 5, Topic C, Lesson 10-13 Mission 5, Topic E, Lesson 20-21, 23
K.NS.1j	Group a collection of up to 100 objects (e.g., counters, pennies, cubes) into sets of ten and count by tens to determine the total (e.g., there are 3 groups of ten and 6 leftovers, 36 total objects).	Mission 5, Topic D, Lesson 15-18
K.NS.2a	Read, write, and identify the numerals 0 through 30.	Mission 1, Topic D, Lesson 12-16 Mission 1, Topic G, Lesson 29-32 Mission 1, Topic H, Lesson 33-35, 37 Mission 4, Topic A, Lesson 1-6 Mission 4, Topic B, Lesson 7-12 Mission 4, Topic E, Lesson 25, 27 Mission 5, Topic A, Lesson 1-4 Mission 5, Topic C, Lesson 10-13
K.NS.2b	Construct a set of objects that corresponds to a given numeral within 30, including an empty set.	Mission 1, Topic C, Lesson 8 Mission 1, Topic E, Lesson 17-22 Mission 1, Topic F, Lesson 24, 26-28 Mission 3, Topic F, Lesson 23-24 Mission 4, Topic B, Lesson 12 Mission 5, Topic A, Lesson 4-5 Mission 5, Topic B, Lesson 6, 9

VIRGINIA'S MATHEMATICS STANDARDS OF LEARNING		ZEARN MATH
Kindergarten Standards		Lessons
K.NS.2c	Determine and write the numeral that corresponds to the total number of objects in a given set of 30 or fewer concrete objects or pictorial models.	Mission 1, Topic C, Lesson 8-11 Mission 1, Topic D, Lesson 12-16 Mission 1, Topic E, Lesson 17-19, 21 Mission 1, Topic F, Lesson 23-28 Mission 1, Topic G, Lesson 29-32 Mission 1, Topic H, Lesson 33-35, 37 Mission 3, Topic F, Lesson 23-24 Mission 4, Topic A, Lesson 1-6 Mission 4, Topic B, Lesson 7-12 Mission 4, Topic E, Lesson 25, 27 Mission 5, Topic A, Lesson 1-5 Mission 5, Topic B, Lesson 6, 9 Mission 5, Topic C, Lesson 10-14 Mission 5, Topic D, Lesson 18-19 Mission 5, Topic E, Lesson 22
K.NS.2d	Given a set of up to 30 objects, construct another set which has more, fewer, or the same number of objects using concrete or pictorial models.	Mission 3, Topic F, Lesson 20, 22-24
K.NS.2e	Given a numeral up to 30, construct a set which has more, fewer, or the same number of objects using concrete or pictorial models.	Mission 3, Topic G, Lesson 25-28
K.NS.2f	Compare two sets containing up to 30 concrete objects or pictorial models, using the terms more, fewer, or the same as (equal to).	Mission 3, Topic E, Lesson 17-19 Mission 3, Topic F, Lesson 21 Mission 3, Topic G, Lesson 25-27
K.NS.2g	Compare numbers up to 30, to the benchmarks of 5 and 10 using various models (e.g., five frames, ten frames, number paths [a prelude to number lines], beaded racks, hands) using the terms greater than, less than, or the same as (equal to).	Mission 3, Topic F, Lesson 20, 23 Mission 3, Topic G, Lesson 25-26, 28
Computation and Estimation		
K.CE.1a	Use objects, drawings, words, or numbers to compose and decompose numbers less than or equal to 5 in multiple ways.	Mission 1, Topic C, Lesson 9-10 Mission 1, Topic D, Lesson 16 Mission 4, Topic A, Lesson 1-6

VIRGINIA'S MATHEMATICS STANDARDS OF LEARNING		ZEARN MATH
Kindergarten Standards		Lessons
K.CE.1b	Recognize and describe with fluency part-part-whole relationships for numbers up to 5 in a variety of configurations.	Mission 1, Topic C, Lesson 9-10 Mission 1, Topic D, Lesson 16 Mission 4, Topic A, Lesson 1-6
K.CE.1c	Model and identify the number that makes 5 when added to a given number less than or equal to 5.	Mission 1, Topic C, Lesson 9-10 Mission 1, Topic D, Lesson 16 Mission 4, Topic A, Lesson 1-6
K.CE.1d	Use objects, drawings, words, or numbers to compose and decompose numbers less than or equal to 10 in multiple ways.	Mission 1, Topic C, Lesson 9-11 Mission 1, Topic D, Lesson 14- 15 Mission 4, Topic A, Lesson 2, 4-6 Mission 4, Topic B, Lesson 7-12 Mission 4, Topic D, Lesson 22-24 Mission 4, Topic E, Lesson 25-28 Mission 4, Topic F, Lesson 32
K.CE.1e	Model and identify the number that makes 10 when added to a given number less than or equal to 10.	Mission 4, Topic H, Lesson 39-40
K.CE.1f	Model and solve single-step contextual problems (join, separate, and part-part-whole) using 10 or fewer concrete objects.	Mission 4, Topic C, Lesson 16-18 Mission 4, Topic D, Lesson 19, 21 Mission 4, Topic F, Lesson 31 Mission 4, Topic G, Lesson 34-36 Mission 4, Topic H, Lesson 37
Measurement and Geometry		
K.MG.1a	Use direct comparisons to compare, describe, and justify the: i. lengths of two objects using the terms longer or shorter; ii. heights of two objects using the terms taller or shorter; iii. weights of two objects using the terms heavier or lighter; iv. volumes of two containers using the terms more or less; and v. amount of time spent on two events using the terms longer or shorter	Mission 3, Topic A, Lesson 1-3 Mission 3, Topic B, Lesson 4-7 Mission 3, Topic C, Lesson 8-12 Mission 3, Topic D, Lesson 13-15 Mission 3, Topic F, Lesson 20 Mission 3, Topic G, Lesson 28 Mission 3, Topic H, Lesson 29-32
K.MG.2a	Identify and name concrete and pictorial representations of circles, triangles, squares, and rectangles regardless of their orientation in space.	Mission 2, Topic A, Lesson 2-5

VIRGINIA'S MATHEMATICS STANDARDS OF LEARNING		ZEARN MATH
Kindergarten Standards		Lessons
K.MG.2b	Describe triangles, squares, and rectangles to include the number of sides and number of vertices.	Mission 2, Topic A, Lesson 1-5
K.MG.2c	Describe a circle using terms such as round and curved.	Mission 2, Topic A, Lesson 4-5
K.MG.2d	Distinguish between examples and nonexamples of identified plane figures (circles, triangles, squares, and rectangles).	Mission 2, Topic A, Lesson 2-4
K.MG.2e	Compare and contrast two plane figures using characteristics to describe similarities and differences.	Mission 2, Topic A, Lesson 1-5
K.MG.2f	Construct plane figures (circles, triangles, squares, and rectangles) using a variety of materials (e.g., straws, sticks, pipe cleaners).	Mission 6, Topic A, Lesson 1-3
K.MG.3a	Identify a calendar as a tool used to measure time.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
K.MG.3b	Name the days of the week and state that there are seven days in one week	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
K.MG.3c	Determine the day before and after a given day (e.g., yesterday, today, tomorrow).	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
K.MG.3d	Name the twelve months of the year and state that there are twelve months in one year.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
K.MG.3e	Distinguish between days of the week and months of the year.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
Probability and Statistics		
K.PS.1a	Sort and classify concrete objects into appropriate subsets (categories) based on one attribute (e.g., size, shape, color, thickness).	Mission 1, Topic A, Lesson 1-3 Mission 1, Topic B, Lesson 4-6 Mission 1, Topic C, Lesson 7

VIRGINIA'S MATHEMATICS STANDARDS OF LEARNING		ZEARN MATH
Kindergarten Standards		Lessons
		Mission 2, Topic A, Lesson 1 Mission 2, Topic B, Lesson 7
K.PS.1b	Describe and label attributes (e.g., size, color, shape) of a set of objects (e.g., coins, counters, buttons) that has been sorted.	Mission 1, Topic B, Lesson 4
K.PS.1c	Pose questions, given a predetermined context, that require the collection of data (limited to 25 or fewer data points for no more than four categories).	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
K.PS.1d	Determine the data needed to answer a posed question, and collect the data using various methods (e.g., counting objects, drawing pictures).	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
K.PS.1e	Organize and represent a data set (vertically or horizontally) by sorting concrete objects into organized groups to form a simple object graph.	Mission 1, Topic D, Lesson 15
K.PS.1f	Organize and represent a data set (vertically or horizontally) using pictures to form a simple picture graph.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
K.PS.1g	Analyze data represented in object graphs and picture graphs and communicate results: i. ask and answer questions about the data represented in object graphs and picture graphs (e.g., how many in each category, which categories have the greatest, least, or the same amount of data); and ii. draw conclusions about the data and make predictions based on the data.	Mission 1, Topic D, Lesson 15
Patterns, Functions, and Algebra		
K.PFA.1a	Identify and describe the core found in repeating patterns.	Mission 5, Topic C, Lesson 11-13
K.PFA.1b	Extend a repeating pattern by adding at least two complete repetitions of the core to the pattern.	Mission 5, Topic C, Lesson 11-13

VIRGINIA'S MATHEMATICS STANDARDS OF LEARNING		ZEARN MATH
Kindergarten Standards		Lessons
K.PFA.1c	Create and describe a repeating pattern using objects, colors, sounds, movements, or pictures.	Mission 5, Topic C, Lesson 11-13

1st Grade

VIRGINIA'S MATHEMATICS STANDARDS OF LEARNING		ZEARN MATH
1st Grade Standards		Lessons
Number and Number Sense		
1.NS.1a	Count forward orally by ones from 0 to 120 starting at any number between 0 and 120.	Mission 4, Topic A, Lesson 1-6 Mission 6, Topic B, Lesson 4, 7, 9
1.NS.1b	Count backward orally by ones when given any number between 1 and 30.	Mission 4, Topic A, Lesson 3 Mission 4, Topic B, Lesson 8 Mission 4, Topic E, Lesson 20
1.NS.1c	Represent forward counting patterns when counting by groups of 5 and groups of 10 up to 120 using a variety of tools (e.g., objects, coins, 120 chart).	Mission 4, Topic A, Lesson 1, 3-4 Mission 4, Topic C, Lesson 12 Mission 4, Topic F, Lesson 23-24
1.NS.1d	Represent forward counting patterns when counting by groups of 2 up to at least 30 using a variety of tools (e.g., beaded number strings, number paths [a prelude to number lines], 120 chart).	Mission 4, Topic A, Lesson 1
1.NS.1e	Group a collection of up to 120 objects into tens and ones, and count to determine the total (e.g., 5 groups of ten and 6 ones is equal to 56 total objects).	Mission 2, Topic D, Lesson 26-29 Mission 4, Topic A, Lesson 1-4, 6 Mission 4, Topic F, Lesson 23 Mission 6, Topic B, Lesson 3-4, 8-9
1.NS.1f	Identify a penny, nickel, and dime by their attributes and describe the number of pennies equivalent to a nickel and a dime.	Mission 4, Topic A, Lesson 6 Mission 6, Topic E, Lesson 20-21
1.NS.1g	Count by ones, fives, or tens to determine the value of a collection of like coins (pennies, nickels, or dimes), whose total value is 100 cents or less.	Mission 4, Topic A, Lesson 3 Mission 4, Topic C, Lesson 12 Mission 4, Topic F, Lesson 23-24
1.NS.2a	Read and write numerals 0-120 in sequence and out of sequence.	Mission 4, Topic A, Lesson 1-6 Mission 6, Topic B, Lesson 4, 7, 9
1.NS.2b	Estimate the number of objects (up to 120) in a given collection and justify the reasonableness of an answer.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>

VIRGINIA'S MATHEMATICS STANDARDS OF LEARNING		ZEARN MATH
1st Grade Standards		Lessons
1.NS.2c	Create a concrete or pictorial representation of a number using tens and ones and write the corresponding numeral up to 120 (e.g., 47 can be represented as 47 ones or it can be grouped into 4 tens with 7 ones left over).	Mission 2, Topic D, Lesson 26-29 Mission 4, Topic A, Lesson 1-4, 6 Mission 4, Topic F, Lesson 23 Mission 6, Topic B, Lesson 3-4, 8-9
1.NS.2d	Describe the number of groups of tens and ones when given a two-digit number and justify reasoning.	Mission 2, Topic D, Lesson 26-29 Mission 4, Topic A, Lesson 1-4, 6 Mission 4, Topic F, Lesson 23 Mission 6, Topic B, Lesson 3-4, 8-9
1.NS.2e	Compare two numbers between 0 and 120 represented pictorially or with concrete objects using the terms greater than, less than, or equal to.	Mission 4, Topic B, Lesson 7-10 Mission 6, Topic B, Lesson 6
1.NS.2f	Order three sets, each set containing up to 120 objects, from least to greatest, and greatest to least.	Mission 4, Topic B, Lesson 8-9
1.NS.3a	Represent equal shares of a whole with two or four sharers, when given a contextual problem.	Mission 5, Topic C, Lesson 8-9 Mission 5, Topic D, Lesson 10
1.NS.3b	Represent and name halves and fourths of a whole, using a region/area model (e.g., pie pieces, pattern blocks, paper folding, drawings) and a set model (e.g., eggs, marbles, counters) limited to two or four items.	Mission 5, Topic C, Lesson 7 Mission 5, Topic D, Lesson 10
1.NS.3c	Describe and justify how shares are equal pieces or equal parts of the whole (limited to halves, fourths) when given a contextual problem.	Mission 5, Topic C, Lesson 8-9 Mission 5, Topic D, Lesson 10
Computation and Estimation		
1.CE.1a	Recognize and describe with fluency part-part-whole relationships for numbers up to 10 in a variety of configurations.	Mission 1, Topic A, Lesson 1-3 Mission 1, Topic B, Lesson 4-8 Mission 1, Topic D, Lesson 14-16 Mission 1, Topic G, Lesson 26-27 Mission 1, Topic I, Lesson 35-37 Mission 1, Topic J, Lesson 38-39
1.CE.1b	Demonstrate fluency with addition and subtraction within 10 by applying reasoning strategies (e.g., count on/count back, one more/one less, doubles, make ten).	Mission 1, Topic A, Lesson 1-3 Mission 1, Topic B, Lesson 4-8 Mission 1, Topic D, Lesson 14-16 Mission 1, Topic G, Lesson 26-27

VIRGINIA'S MATHEMATICS STANDARDS OF LEARNING		ZEARN MATH
1st Grade Standards		Lessons
		Mission 1, Topic F, Lesson 21-24 Mission 1, Topic I, Lesson 33-37 Mission 1, Topic J, Lesson 38-39 Mission 2, Topic A, Lesson 1-11
1.CE.1c	Recall with automaticity addition and subtraction facts within 10.	Mission 1, Topic A, Lesson 1-3 Mission 1, Topic B, Lesson 4-8 Mission 1, Topic D, Lesson 14-16 Mission 1, Topic G, Lesson 26-27 Mission 1, Topic F, Lesson 21-24 Mission 1, Topic I, Lesson 33-37 Mission 1, Topic J, Lesson 38-39 Mission 2, Topic A, Lesson 1-11
1.CE.1d	Investigate, recognize, and describe part-part-whole relationships for numbers up to 20 in a variety of configurations (e.g., beaded racks, double ten frames).	Mission 1, Topic C, Lesson 9-13 Mission 1, Topic D, Lesson 14-16 Mission 1, Topic E, Lesson 17-20 Mission 1, Topic F, Lesson 21-24 Mission 1, Topic G, Lesson 26-27 Mission 1, Topic I, Lesson 33-37 Mission 1, Topic J, Lesson 38-39 Mission 2, Topic A, Lesson 1-11 Mission 2, Topic B, Lesson 12-21 Mission 2, Topic C, Lesson 25 Mission 2, Topic D, Lesson 27-29
1.CE.1e	Solve addition and subtraction problems within 20 using various strategies (e.g., inverse relationships: if $9 + 3 = 12$ then $12 - 3 = 9$; decomposition using known sums/differences: $9 + 7$ can be thought of as 9 decomposed into 2 and 7, then use doubles, $7 + 7 = 14$; $14 + 2 = 16$ or decompose the 7 into 1 and 6; make a ten: $1 + 9 = 10$; $10 + 6 = 16$).	Mission 1, Topic C, Lesson 9-13 Mission 1, Topic D, Lesson 14-16 Mission 1, Topic E, Lesson 17-20 Mission 1, Topic F, Lesson 21-24 Mission 1, Topic G, Lesson 26-27 Mission 1, Topic I, Lesson 33-37 Mission 1, Topic J, Lesson 38-39 Mission 2, Topic A, Lesson 1-11 Mission 2, Topic B, Lesson 12-21 Mission 2, Topic C, Lesson 25 Mission 2, Topic D, Lesson 27-29
1.CE.1f	Represent, solve, and justify solutions to single-step addition and subtraction problems (join, separate, and part-part-whole) within 20, including those in context, using words, objects, drawings, or numbers.	Mission 1, Topic B, Lesson 4-8 Mission 1, Topic C, Lesson 9-13 Mission 1, Topic G, Lesson 25 Mission 1, Topic H, Lesson 28-32 Mission 2, Topic A, Lesson 3-5, 7-9, 11

VIRGINIA'S MATHEMATICS STANDARDS OF LEARNING		ZEARN MATH
1st Grade Standards		Lessons
		Mission 2, Topic B, Lesson 12-13, 17-18, 21 Mission 2, Topic C, Lesson 22-24 Mission 2, Topic D, Lesson 27-29 Mission 3, Topic D, Lesson 10-13 Mission 4, Topic E, Lesson 19-22 Mission 6, Topic A, Lesson 1-2 Mission 6, Topic F, Lesson 25-27
1.CE.1g	Determine the unknown whole number that will result in a sum or difference of 10 or 20 (e.g., $14 - \underline{\quad} = 10$ or $15 + \underline{\quad} = 20$).	Mission 1, Topic D, Lesson 16 Mission 1, Topic H, Lesson 29-30, 32
1.CE.1h	Identify and use (+) as a symbol for addition and (-) as a symbol for subtraction.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
1.CE.1i	Describe the equal symbol (=) as a balance representing an equivalent relationship between expressions on either side of the equal symbol (e.g., 6 and 1 is the same as 4 and 3; $6 + 1$ is balanced with $4 + 3$; $6 + 1 = 4 + 3$).	Mission 1, Topic E, Lesson 17-19 Mission 2, Topic C, Lesson 25
1.CE.1j	Use concrete materials to model, identify, and justify when two expressions are not equal (e.g., $10 - 3$ is not equal to $3 + 5$).	Mission 1, Topic E, Lesson 17-19 Mission 2, Topic C, Lesson 25
1.CE.1k	Use concrete materials to model an equation that represents the relationship of two expressions of equal value.	Mission 1, Topic E, Lesson 17-19 Mission 2, Topic C, Lesson 25
1.CE.1l	Write an equation that could be used to represent the solution to an oral, written, or picture problem.	Mission 1, Topic C, Lesson 9-13 Mission 1, Topic H, Lesson 29
Measurement and Geometry		
1.MG.1a	Use nonstandard units to measure the: - lengths of two objects (units laid end to end with no gaps or overlaps) and compare the measurements using the terms longer/shorter, taller/shorter, or the same as; - weights of two objects (using a balance scale or a pan scale) and compare the	Mission 3, Topic B, Lesson 4-6 Mission 3, Topic C, Lesson 7-9

VIRGINIA'S MATHEMATICS STANDARDS OF LEARNING		ZEARN MATH
1st Grade Standards		Lessons
	measurements using the terms lighter, heavier, or the same as; and iii. volumes of two containers and compare the measurements using the terms more, less, or the same as.	
1.MG.1b	Measure the length, weight, or volume of the same object or container with two different units and describe how and why the measurements differ.	Mission 2, Topic C, Lesson 7 Mission 7, Topic D, Lesson 18
1.MG.2a	Describe triangles, squares, and rectangles using the terms sides, vertices, and angles. Describe a circle using terms such as <i>round</i> and <i>curved</i> .	Mission 5, Topic A, Lesson 1-2
1.MG.2b	Sort plane figures based on their characteristics (e.g., number of sides, vertices, angles, curved).	Mission 5, Topic A, Lesson 1-2
1.MG.2c	Draw and name the plane figure (circle, square, rectangle, triangle) when given information about the number of sides, vertices, and angles.	Mission 5, Topic A, Lesson 1
1.MG.2d	Identify, name, and describe representations of circles, squares, rectangles, and triangles, regardless of orientation, in different environments and explain reasoning.	Kindergarten, Mission 2, Topic A, Lesson 2-5 Kindergarten, Mission 2, Topic B, Lesson 7-8 Kindergarten, Mission 2, Topic C, Lesson 9-10
1.MG.2e	Recognize and name the angles found in rectangles and squares as right angles.	Mission 5, Topic A, Lesson 2
1.MG.2f	Compose larger plane figures by combining two or three simple plane figures (triangles, squares, and/or rectangles).	Mission 5, Topic B, Lesson 4-6
1.MG.3a	Identify different tools to measure time including clocks (analog and digital) and calendar.	Mission 5, Topic D, Lesson 10, 13
1.MG.3b	Describe the units of time represented on a clock as minutes and hours.	Mission 5, Topic D, Lesson 10-11

VIRGINIA'S MATHEMATICS STANDARDS OF LEARNING		ZEARN MATH
1st Grade Standards		Lessons
1.MG.3c	Tell time to the hour and half-hour, using analog and digital clocks.	Mission 5, Topic D, Lesson 10
1.MG.3d	Describe the location of the hour hand relative to time to the hour and half-hour on an analog clock	Mission 5, Topic D, Lesson 11-13
1.MG.3e	Describe the location of the minute hand relative to time to the hour and half-hour on an analog clock.	Mission 5, Topic D, Lesson 11-13
1.MG.3f	Match the time shown on a digital clock to an analog clock to the hour and half-hour.	Mission 5, Topic D, Lesson 10
1.MG.3g	Identify specific days/dates on a calendar (e.g., What date is Saturday? How many Fridays are in October?)	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
1.MG.3h	Use ordinal numbers first through tenth to describe the relative position of specific days/dates (e.g., What is the first Monday in October? What day of the week is May 6th?).	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
1.MG.3i	Determine the day/date before and after a given day/date (e.g., Today is the 8 th , so yesterday was the ?), and a date that is a specific number of days/weeks in the past or future (e.g., Tim's birthday is in 10 days, what will be the date of his birthday?).	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
Probability and Statistics		
1.PS.1a	Sort and classify concrete objects into appropriate subsets (categories) based on one or two attributes, such as size, shape, color, and/or thickness (e.g., sort a set of objects that are both red and thick).	Mission 3, Topic D, Lesson 10-13 <i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
1.PS.1b	Describe and label attributes of a set of objects that has been sorted.	Mission 3, Topic D, Lesson 10-13 <i>It is recommended that teachers use supplemental materials to fully address this standard.</i>

VIRGINIA'S MATHEMATICS STANDARDS OF LEARNING		ZEARN MATH
1st Grade Standards		Lessons
1.PS.1c	Pose questions, given a predetermined context, that require the collection of data (limited to 25 or fewer data points for no more than four categories).	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
1.PS.1d	Determine the data needed to answer a posed question and collect the data using various methods (e.g., counting objects, drawing pictures, tallying).	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
1.PS.1e	Organize and represent a data set by sorting the collected data using various methods (e.g., tallying, T-charts).	Mission 3, Topic D, Lesson 10-13 <i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
1.PS.1f	Represent a data set (vertically or horizontally) using object graphs, picture graphs, and tables.	Mission 3, Topic D, Lesson 10-13 <i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
1.PS.1g	Analyze data represented in object graphs, picture graphs, and tables and communicate results: i. ask and answer questions about the data represented in object graphs, picture graphs, and tables (e.g., total number of data points represented, how many in each category, how many more or less are in one category than another); and ii. draw conclusions about the data and make predictions based on the data.	Mission 3, Topic D, Lesson 10-13 <i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
Patterns, Functions, and Algebra		
1.PFA.1a	Identify and describe repeating and increasing patterns.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
1.PFA.1b	Analyze a repeating or increasing pattern and generalize the change to extend the pattern using objects, colors, movements, pictures, or geometric figures.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>

VIRGINIA'S MATHEMATICS STANDARDS OF LEARNING		ZEARN MATH
1st Grade Standards		Lessons
1.PFA.1c	Create a repeating or increasing pattern using objects, pictures, movements, colors, or geometric figures.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
1.PFA.1d	Transfer a repeating or increasing pattern from one form to another.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>

2nd Grade

VIRGINIA'S MATHEMATICS STANDARDS OF LEARNING		ZEARN MATH
2nd Grade Standards		Lessons
Number and Number Sense		
2.NS.1a	Represent forward counting patterns when counting by groups of 2 up to at least 50, starting at various multiples of 2 and using a variety of tools (e.g., objects, number lines, hundreds charts).	Mission 6, Topic D, Lesson 17-19
2.NS.1b	Represent forward counting patterns created when counting by groups of 5s, 10s, and 25s starting at various multiples up to at least 200 using a variety of tools (e.g., objects, number lines, hundreds charts).	Mission 3, Topic B, Lesson 2-3 Mission 3, Topic C, Lesson 4 Mission 3, Topic D, Lesson 8-9 Mission 3, Topic E, Lesson 12-15 Mission 3, Topic G, Lesson 19-21 Mission 7, Topic E, Lesson 21 Mission 8, Topic D, Lesson 14
2.NS.1c	Describe and use patterns in skip counting by multiples of 2 (to at least 50), and multiples of 5, 10, and 25 (to at least 200) to justify the next number in the counting sequence.	Mission 3, Topic B, Lesson 2-3 Mission 3, Topic C, Lesson 4 Mission 3, Topic D, Lesson 8-9 Mission 3, Topic E, Lesson 12-15 Mission 3, Topic G, Lesson 19-21 Mission 7, Topic E, Lesson 21 Mission 8, Topic D, Lesson 14
2.NS.1d	Represent forward counting patterns when counting by groups of 100 up to at least 1,000 starting at 0 using a variety of tools (e.g., objects, number lines, calculators, one thousand charts).	Mission 3, Topic B, Lesson 3 Mission 3, Topic C, Lesson 4 Mission 3, Topic D, Lesson 8-9 Mission 3, Topic G, Lesson 19-21
2.NS.1e	Represent backward counting patterns when counting by groups of 10 from 200 or less using a variety of tools including objects, number lines, calculators, and hundreds charts.	Mission 3, Topic E, Lesson 11 Mission 3, Topic G, Lesson 20-21
2.NS.1f	Describe and use patterns in skip counting backwards by 10s (from at least 200) to justify the next number in the counting sequence.	Mission 3, Topic E, Lesson 11 Mission 3, Topic G, Lesson 20-21

VIRGINIA'S MATHEMATICS STANDARDS OF LEARNING		ZEARN MATH
2nd Grade Standards		Lessons
2.NS.1g	Choose a reasonable estimate up to 1,000 when given a contextual problem (e.g., What would be the best estimate for the number of students in our school – 5, 50, or 500?).	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
2.NS.1h	Represent even numbers (up to 50) with concrete objects, using two equal groups or two equal addends.	Mission 6, Topic D, Lesson 17-19
2.NS.1i	Represent odd numbers (up to 50) with concrete objects, using two equal groups with one leftover or two equal addends plus 1.	Mission 6, Topic D, Lesson 17-19
2.NS.1j	Determine whether a number (up to 50) is even or odd using concrete objects and justify reasoning (e.g., dividing collections of objects into two equal groups, pairing objects).	Mission 6, Topic D, Lesson 17-19
2.NS.2a	Write the three-digit whole number represented by a given model (e.g., concrete objects, pictures of base 10 blocks).	Mission 3, Topic C, Lesson 4-7 Mission 3, Topic D, Lesson 8-9 Mission 3, Topic E, Lesson 13, 14
2.NS.2b	Read, write, and represent three-digit numbers in standard form, expanded form, and word form, using concrete or pictorial representations.	Mission 3, Topic C, Lesson 4-7 Mission 3, Topic D, Lesson 8-9 Mission 3, Topic E, Lesson 13, 14
2.NS.2c	Apply patterns within the base 10 system to determine and communicate, orally and in written form, the place (ones, tens, hundreds) and value of each digit in a three-digit whole number (e.g., in 352, the 5 represents 5 tens and its value is 50).	Mission 3, Topic A, Lesson 1 Mission 3, Topic B, Lesson 2-3 Mission 3, Topic C, Lesson 4-7 Mission 3, Topic D, Lesson 8-9 Mission 3, Topic E, Lesson 11-15
2.NS.2d	Investigate and explain the ten-to-one relationships among ones, tens, and hundreds, using models.	Mission 3, Topic A, Lesson 1 Mission 3, Topic B, Lesson 2 Mission 3, Topic C, Lesson 4 Mission 3, Topic E, Lesson 12
2.NS.2e	Compose and decompose whole numbers up to 200 by making connections between a variety of models (e.g., base 10 blocks, place value cards, presented orally, in expanded or standard form) and counting strategies (e.g., 156 can be 1 hundred, 5 tens, 6 ones; 1 hundred, 4 tens, 16 ones; 15 tens, 6 ones).	Mission 3, Topic A, Lesson 1 Mission 3, Topic C, Lesson 4-7 Mission 3, Topic D, Lesson 8-9 Mission 3, Topic E, Lesson 11-15

VIRGINIA'S MATHEMATICS STANDARDS OF LEARNING		ZEARN MATH
2nd Grade Standards		Lessons
2.NS.2f	Plot and justify the position of a given number up to 100 on a number line with pre-marked benchmarks of 1s, 2s, 5s, 10s, or 25s.	Mission 2, Topic D, Lesson 8 Mission 7, Topic A, Lesson 3 Mission 7, Topic E, Lesson 21-22 Mission 7, Topic F, Lesson 24
2.NS.2g	Compare two whole numbers, each 999 or less, represented concretely, pictorially, or symbolically, using words (greater than, less than, or equal to) and symbols ($>$, $<$, or $=$). Justify reasoning orally, in writing, or with a model.	Mission 3, Topic F, Lesson 16-18
2.NS.2h	Order up to three whole numbers, each 999 or less, represented concretely, pictorially, or symbolically from least to greatest and greatest to least.	Mission 3, Topic F, Lesson 18
2.NS.3a	Model and describe fractions as representing equal-size parts of a whole.	Mission 8, Topic B, Lesson 7-8 Mission 8, Topic C, Lesson 9-12 Mission 8, Topic D, Lesson 13
2.NS.3b	Describe the relationship between the number of fractional parts needed to make a whole and the size of the parts (i.e., as the whole is divided into more parts, each part becomes smaller).	Grade 1, Mission 5, Topic C, Lesson 9
2.NS.3c	Compose the whole for a given fractional part and its value (in context) for halves, fourths, eighths, thirds, and sixths (e.g., when given $\frac{1}{4}$, determine how many pieces would be needed to make $\frac{4}{4}$).	Mission 8, Topic C, Lesson 11
2.NS.3d	Using same-size fraction pieces, from a region/area model, count by unit fractions up to two wholes (e.g., zero one-fourths, one one-fourth, two one-fourths, three one-fourths, four one-fourths, five one-fourths; or zero-fourths, one-fourth, two-fourths, three-fourths, four-fourths, five-fourths).	Mission 8, Topic C, Lesson 10
2.NS.3e	Given a context, represent, name, and write fractional parts of a whole for halves, fourths, eighths, thirds, and sixths using: i. region/area models (e.g., pie pieces, pattern blocks, geoboards); ii. length models (e.g., paper fraction strips, fraction bars, rods, number lines); and	Grade 3, Mission 5, Topic A, Lesson 1-4 Grade 3, Mission 5, Topic B, Lesson 5-9

VIRGINIA'S MATHEMATICS STANDARDS OF LEARNING		ZEARN MATH
2nd Grade Standards		Lessons
	iii. set models (e.g., chips, counters, cubes).	
2.NS.3f	Compare unit fractions for halves, fourths, eighths, thirds, and sixths using words (greater than, less than or equal to) and symbols ($>$, $<$, $=$), with region/area and length models.	Grade 3, Mission 5, Topic C, Lesson 10-11, 13
2.NS.4a	Identify a quarter and its value and determine multiple ways to represent the value of a quarter using pennies, nickels, and/or dimes.	Mission 7, Topic B, Lesson 9-10
2.NS.4b	Count by ones, fives, tens, and twenty-fives to determine the value of a collection of mixed coins and one-dollar bills whose total value is \$2.00 or less.	Mission 3, Topic D, Lesson 10 Mission 7, Topic B, Lesson 6-7, 9-10, 12-13
2.NS.4c	Construct a set of coins and/or bills to total a given amount of money whose value is \$2.00 or less.	Mission 7, Topic B, Lesson 9-10
2.NS.4d	Represent the value of a collection of coins and one-dollar bills (limited to \$2.00 or less) using the cent (¢) and dollar (\$) symbols and decimal point (.).	Mission 7, Topic B, Lesson 6-7, 9-10, 12-13
Computation and Estimation		
2.CE.1a	Apply strategies (e.g., rounding to the nearest 10, compatible numbers, other number relationships) to estimate a solution for single-step addition or subtraction problems, including those in context, where addends and minuends do not exceed 100.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
2.CE.1b	Apply strategies (e.g., the use of concrete and pictorial models, place value, properties of addition, the relationship between addition and subtraction) to determine the sum or difference of two whole numbers where addends or minuends do not exceed 100.	Mission 1, Topic B, Lesson 3, 5-6, 8 Mission 4, Topic A, Lesson 2-4 Mission 4, Topic B, Lesson 6-8, 10 Mission 4, Topic C, Lesson 11-13 Mission 7, Topic B, Lesson 7-8, 11-13
2.CE.1c	Apply strategies (e.g., the use of concrete and pictorial models, place value, properties of addition, the relationship between addition and subtraction) to determine the sum or difference of two whole numbers where addends or minuends do not exceed 100.	Mission 1, Topic B, Lesson 3, 5-6, 8 Mission 4, Topic A, Lesson 1-5 Mission 4, Topic B, Lesson 6-10 Mission 4, Topic C, Lesson 11-14, 16 Mission 4, Topic F, Lesson 31

VIRGINIA'S MATHEMATICS STANDARDS OF LEARNING		ZEARN MATH
2nd Grade Standards		Lessons
		Mission 7, Topic B, Lesson 7-8, 12-13
2.CE.1d	Demonstrate fluency with addition and subtraction within 20 by applying reasoning strategies (e.g., doubles, near doubles, make-a-ten, compensations, inverse relationships).	Mission 1, Topic A, Lesson 1-2 Mission 1, Topic B, Lesson 4, 7
2.CE.1e	Recall with automaticity addition and subtraction facts within 20.	Mission 1, Topic A, Lesson 1-2 Mission 1, Topic B, Lesson 4, 7
2.CE.1f	Use patterns, models, and strategies to make generalizations about the algebraic properties for fluency (e.g., $4 + 3$ is equal to $3 + 4$; $0 + 8 = 8$).	Grade 1, Mission 1, Topic E, Lesson 17-19 Grade 1, Mission 2, Topic A, Lesson 6, 10 Grade 1, Mission 2, Topic B, Lesson 20 Grade 1, Mission 2, Topic C, Lesson 25
2.CE.1g	Determine the missing number in an equation (number sentence) through modeling and justification with addition and subtraction within 20 (e.g., $3 + \square = 5$ or $+ 2 = 5$; $5 - \square = 3$ or $5 - 2 = \square$).	Grade 1, Mission 1, Topic D, Lesson 16 Grade 1, Mission 1, Topic H, Lesson 29-30, 32
2.CE.1h	Use inverse relationships to write all related facts connected to a given addition or subtraction fact model within 20 (e.g., given a model for $3 + 4 = 7$, write $4 + 3 = 7$, $7 - 4 = 3$, and $7 - 3 = 4$).	Mission 1, Topic A, Lesson 1-2 Mission 1, Topic B, Lesson 4, 7
2.CE.1i	Describe the not equal symbol ($\neq$) as representing a relationship where expressions on either side of the not equal symbol represent different values and justify reasoning.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
2.CE.1j	Represent and justify the relationship between values and expressions as equal or not equal using appropriate models and/or symbols (e.g., $9 + 24 = 10 + 23$; $45 - 9 = 46 - 10$; $15 + 16 \neq 31 + 15$).	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
Measurement and Geometry		

VIRGINIA'S MATHEMATICS STANDARDS OF LEARNING		ZEARN MATH
2nd Grade Standards		Lessons
2.MG.1a	Explain the purpose of various measurement tools and how to use them appropriately by: i. identifying a ruler as an instrument to measure length; ii. identifying different types of scales as instruments to measure weight; and iii. identifying different types of measuring cups as instruments to measure liquid volume.	Mission 2, Topic A, Lesson 1-3
2.MG.1b	Use U.S. Customary units to estimate, measure, and compare the two for reasonableness: i. the length of an object to the nearest inch, using a ruler; ii. the weight of an object to the nearest pound, using a scale; and iii. the liquid volume of a container to the nearest cup, using a measuring cup.	Mission 2, Topic B, Lesson 4-5 Mission 2, Topic C, Lesson 6-7 Mission 2, Topic D, Lesson 9 Mission 7, Topic D, Lesson 17, 19
2.MG.2a	Identify the number of minutes in an hour (60 minutes) and the number of hours in a day (24 hours).	Mission 8, Topic D, Lesson 13-14
2.MG.2b	Determine the unit of time (minutes, hours, days, or weeks) that is most appropriate when measuring a given activity or context and explain reasoning (e.g., Would you measure the time it takes to brush your teeth in minutes or hours?).	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
2.MG.2c	Show, tell, and write time to the nearest five minutes, using analog and digital clocks.	Mission 8, Topic D, Lesson 13-15
2.MG.2d	Match a written time (e.g., 1:35, 6:20, 9:05) to the time shown on an analog clock to the nearest five minutes.	Mission 8, Topic D, Lesson 13-15
2.MG.3a	Explore a figure using a variety of tools (e.g., paper folding, geoboards, drawings) to show and justify a line of symmetry, if one exists.	Grade 4, Mission 4, Topic D, Lesson 12
2.MG.3b	Create figures with at least one line of symmetry using various concrete and pictorial representations.	Grade 4, Mission 4, Topic D, Lesson 12
2.MG.3c	Describe the two resulting figures formed by a line of symmetry as being congruent (having the same shape and size).	Grade 4, Mission 4, Topic D, Lesson 12

VIRGINIA'S MATHEMATICS STANDARDS OF LEARNING		ZEARN MATH
2nd Grade Standards		Lessons
2.MG.4a	Trace faces of solid figures (cubes and rectangular prisms) to create the set of plane figures related to the solid figure.	Mission 8, Topic A, Lesson 5
2.MG.4b	Compare and contrast models and nets (cutouts) of cubes and rectangular prisms (e.g., number and shapes of faces, edges, vertices).	Mission 8, Topic A, Lesson 5
2.MG.4c	Given a concrete or pictorial model, name and describe the solid figure (sphere, cube, and rectangular prism) by its characteristics (e.g., number of edges, number of vertices, shapes of faces).	Mission 8, Topic A, Lesson 1-5 Mission 8, Topic B, Lesson 6
2.MG.4d	Compare and contrast plane and solid figures (circles/spheres, squares/cubes, and rectangles/rectangular prisms) according to their characteristics (e.g., number and shapes of their faces, edges, vertices).	Mission 8, Topic A, Lesson 5
Probability and Statistics		
2.PS.1a	Pose questions, given a predetermined context, that require the collection of data (limited to 25 or fewer data points for no more than six categories).	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
2.PS.1b	Determine the data needed to answer a posed question and collect the data using various methods (e.g., voting; creating lists, tables, or charts; tallying).	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
2.PS.1c	Organize and represent a data set using a pictograph where each symbol represents up to 2 data points. Determine and use a key to assist in the analysis of the data.	Mission 7, Topic A, Lesson 1-2
2.PS.1d	Organize and represent a data set using a bar graph with a title and labeled axes (limited to 25 or fewer data points for up to six categories, and limit increments of scale to multiples of 1 or 2).	Mission 7, Topic A, Lesson 3-5
2.PS.1e	Analyze data represented in pictographs and bar graphs and communicate results: i. ask and answer questions about the data represented in pictographs and bar graphs	Mission 7, Topic A, Lesson 1-5

VIRGINIA'S MATHEMATICS STANDARDS OF LEARNING		ZEARN MATH
2nd Grade Standards		Lessons
	(e.g., total number of data points represented, how many in each category, how many more or less are in one category than another). Pictograph keys will be limited to symbols representing 1, 2, 5, or 10 pieces of data and bar graphs will be limited to scales with increments in multiples of 1, 2, 5, or 10; and ii. draw conclusions about the data and make predictions based on the data.	
Patterns, Functions, and Algebra		
2.PFA.1a	Identify and describe repeating and increasing patterns.	Mission 3, Topic G, Lesson 21 Mission 4, Topic A, Lesson 1 Mission 5, Topic A, Lesson 1 Mission 6, Topic C, Lesson 16 <i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
2.PFA.1b	Analyze a repeating or increasing pattern and generalize the change to extend the pattern using objects, pictures, and numbers.	Mission 3, Topic G, Lesson 21 Mission 4, Topic A, Lesson 1 Mission 5, Topic A, Lesson 1 Mission 6, Topic C, Lesson 16 <i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
2.PFA.1c	Create a repeating or increasing pattern using various representations (e.g., objects, pictures, numbers).	Mission 3, Topic G, Lesson 21 Mission 4, Topic A, Lesson 1 Mission 5, Topic A, Lesson 1 Mission 6, Topic C, Lesson 16 <i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
2.PFA.1d	Transfer a given repeating or increasing pattern from one form to another (e.g., objects, pictures, numbers) and explain the connection between the two patterns.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>

3rd Grade

VIRGINIA'S MATHEMATICS STANDARDS OF LEARNING		ZEARN MATH
3rd Grade Standards		Lessons
Number and Number Sense		
3.NS.1a	Read and write six-digit whole numbers in standard form, expanded form, and word form.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
3.NS.1b	Apply patterns within the base 10 system to determine and communicate, orally and in written form, the place and value of each digit in a six-digit whole number (e.g., in 165,724, the 5 represents 5 thousands and its value is 5,000).	Mission 2, Topic B, Lesson 10-11 <i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
3.NS.1c	Compose, decompose, and represent numbers up to 9,999 in multiple ways, according to place value (e.g., 256 can be 1 hundred, 14 tens, 16 ones, but also 25 tens, 6 ones), with and without models.	Mission 2, Topic D, Lesson 15-17 <i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
3.NS.2a	Compare two whole numbers, each 9,999 or less, using symbols ($>$, $<$, $=$, $\neq$) and/or words (greater than, less than, equal to, not equal to), with and without models.	Grade 4, Mission 1, Topic A, Lesson 3 Grade 4, Mission 1, Topic B, Lesson 4-6
3.NS.2b	Order up to three whole numbers, each 9,999 or less, represented with and without models, from least to greatest and greatest to least.	Grade 4, Mission 1, Topic A, Lesson 3 Grade 4, Mission 1, Topic B, Lesson 4-6
3.NS.3a	Represent, name, and write a given fraction (proper or improper) or mixed number with denominators of 2, 3, 4, 5, 6, 8, and 10 using: - region/area models (e.g., pie pieces, pattern blocks, geoboards); - length models (e.g., paper fraction strips, fraction bars, rods, number lines); and - set models (e.g., chips, counters, cubes).	Mission 5, Topic B, Lesson 5-9 Mission 5, Topic C, Lesson 10, 12-13 Mission 5, Topic E, Lesson 20, 22, 24-25, 27 Mission 5, Topic F, Lesson 28-29

VIRGINIA'S MATHEMATICS STANDARDS OF LEARNING		ZEARN MATH
3rd Grade Standards		Lessons
3.NS.3b	Identify a fraction represented by a model as the sum of unit fractions.	Mission 5, Topic B, Lesson 5-9 Mission 5, Topic C, Lesson 12
3.NS.3c	Use a model of a fraction greater than one to count the fractional parts to name and write it as an improper fraction and as a mixed number (e.g., $\frac{1}{4}$, $\frac{2}{4}$, $\frac{3}{4}$, $\frac{4}{4}$, $\frac{5}{4} = 1\frac{1}{4}$).	Mission 5, Topic B, Lesson 9 Mission 5, Topic D, Lesson 16 Mission 5, Topic E, Lesson 26
3.NS.3d	Compose and decompose fractions (proper and improper) with denominators of 2, 3, 4, 5, 6, 8, and 10 in multiple ways (e.g., $\frac{7}{4} = \frac{4}{4} + \frac{3}{4}$ or $\frac{4}{6} = \frac{3}{6} + \frac{1}{6} = \frac{2}{6} + \frac{2}{6}$) with models.	Grade 4, Mission 5, Topic A, Lesson 1-6 Grade 4, Mission 5, Topic D, Lesson 16-21 Grade 4, Mission 5, Topic E, Lesson 22-25 Grade 4, Mission 5, Topic F, Lesson 29-34
3.NS.3e	Compare a fraction, less than or equal to one, to the benchmarks of 0, $\frac{1}{2}$, and 1 using area/region models, length models, and without models.	Mission 5, Topic C, Lesson 10-11, 13 Mission 5, Topic D, Lesson 16-19 Mission 5, Topic F, Lesson 28-29
3.NS.3f	Compare two fractions (proper or improper) and/or mixed numbers with like numerators of 2, 3, 4, 5, 6, 8, and 10 (e.g., $\frac{2}{3} > \frac{2}{8}$) using words (greater than, less than, equal to) and/or symbols ($>$, $<$, $=$), using area/region models, length models, and without models.	Mission 5, Topic C, Lesson 10-11, 13 Mission 5, Topic D, Lesson 16-19 Mission 5, Topic F, Lesson 28-29
3.NS.3g	Compare two fractions (proper or improper) and/or mixed numbers with like denominators of 2, 3, 4, 5, 6, 8, and 10 (e.g., $\frac{3}{6} < \frac{4}{6}$) using words (greater than, less than, equal to) and/or symbols ($>$, $<$, $=$), using area/region models, length models, and without models.	Mission 5, Topic C, Lesson 10-11, 13 Mission 5, Topic D, Lesson 16-19 Mission 5, Topic F, Lesson 28-29
3.NS.3h	Represent equivalent fractions with denominators of 2, 3, 4, 5, 6, 8, or 10, using region/area models and length models.	Mission 5, Topic E, Lesson 20-27
3.NS.4a	Determine the value of a collection of bills and coins whose total is \$5.00 or less.	Grade 2, Mission 3, Topic D, Lesson 10

VIRGINIA'S MATHEMATICS STANDARDS OF LEARNING		ZEARN MATH
3rd Grade Standards		Lessons
		Grade 2, Mission 7, Topic B, Lesson 6-10, 12-13
3.NS.4b	Construct a set of bills and coins to total a given amount of money whose value is \$5.00 or less.	Grade 2, Mission 3, Topic D, Lesson 10 Grade 2, Mission 7, Topic B, Lesson 6-10, 12-13
3.NS.4c	Compare the values of two sets of coins or two sets of bills and coins, up to \$5.00, with words (greater than, less than, equal to) and/or symbols ($>$, $<$, $=$) using concrete or pictorial models.	Grade 2, Mission 3, Topic D, Lesson 10 Grade 2, Mission 7, Topic B, Lesson 6-10, 12-13
3.NS.4d	Solve contextual problems to make change from \$5.00 or less by using counting on or counting back strategies with concrete or pictorial models.	Grade 2, Mission 3, Topic D, Lesson 10 Grade 2, Mission 7, Topic B, Lesson 6-10, 12-13
Computation and Estimation		
3.CE.1a	Determine and justify whether an estimate or an exact answer is appropriate when solving single-step and multistep contextual problems involving addition and subtraction, where addends and minuends do not exceed 1,000.	Mission 1, Topic F, Lesson 21 Mission 2, Topic D, Lesson 15-17 Mission 2, Topic E, Lesson 18-21 Mission 3, Topic E, Lesson 18 Mission 7, Topic A, Lesson 1-3
3.CE.1b	Apply strategies (e.g., rounding to the nearest 10 or 100, using compatible numbers, using other number relationships) to estimate a solution for single-step or multi step addition or subtraction problems, including those in context, where addends or minuends do not exceed 1,000.	Mission 1, Topic F, Lesson 21 Mission 2, Topic D, Lesson 15-17 Mission 2, Topic E, Lesson 18-21 Mission 3, Topic E, Lesson 18 Mission 7, Topic A, Lesson 1-3
3.CE.1c	Apply strategies (e.g., place value, properties of addition, other number relationships) and algorithms, including the standard algorithm, to determine the sum or difference of two whole numbers where addends and minuends do not exceed 1,000.	Mission 2, Topic A, Lesson 4-5 Mission 2, Topic B, Lesson 10-11 Mission 2, Topic D, Lesson 15-17 Mission 2, Topic E, Lesson 18-21
3.CE.1d	Identify and use the appropriate symbol to distinguish between expressions that are equal and expressions that are not equal (e.g., $256 - 13 = 220 + 23$; $457 + 100 \neq 557 + 100$).	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>

VIRGINIA'S MATHEMATICS STANDARDS OF LEARNING		ZEARN MATH
3rd Grade Standards		Lessons
3.CE.1e	Represent, solve, and justify solutions to single-step and multistep contextual problems involving addition and subtraction with whole numbers where addends and minuends do not exceed 1,000.	Mission 1, Topic F, Lesson 20-21 Mission 3, Topic E, Lesson 18 Mission 3, Topic F, Lesson 21 Mission 7, Topic A, Lesson 1-3
3.CE.2a	Represent multiplication and division of whole numbers through 10×10 , including in a contextual situation, using a variety of approaches and models (e.g., repeated addition/subtraction, equal-sized groups/sharing, arrays, equal jumps on a number line, using multiples to skip count).	Mission 1, Topic A, Lesson 1-3 Mission 1, Topic B, Lesson 4-6 Mission 1, Topic C, Lesson 7-10 Mission 1, Topic D, Lesson 11-13 Mission 1, Topic E, Lesson 14-17 Mission 1, Topic F, Lesson 18 Mission 3, Topic A, Lesson 2-3 Mission 3, Topic B, Lesson 7 Mission 3, Topic C, Lesson 11 Mission 3, Topic D, Lesson 12, 15 Mission 3, Topic E, Lesson 16
3.CE.2b	Use inverse relationships to write the related facts connected to a given model for multiplication and division of whole numbers through 10×10 .	Mission 1, Topic C, Lesson 7-10 Mission 1, Topic E, Lesson 15-16 Mission 1, Topic F, Lesson 18-19 Mission 3, Topic A, Lesson 1-2 Mission 3, Topic B, Lesson 4-6 Mission 3, Topic C, Lesson 9-10 Mission 3, Topic D, Lesson 12-14 Mission 3, Topic E, Lesson 16-17 Mission 3, Topic F, Lesson 20
3.CE.2c	Apply strategies (e.g., place value, the properties of multiplication and/or addition) when multiplying and dividing whole numbers.	Mission 1, Topic C, Lesson 7-10 Mission 1, Topic E, Lesson 15-16 Mission 1, Topic F, Lesson 18-19 Mission 3, Topic A, Lesson 1-2 Mission 3, Topic B, Lesson 4-6 Mission 3, Topic C, Lesson 9-10 Mission 3, Topic D, Lesson 12-14 Mission 3, Topic E, Lesson 16-17 Mission 3, Topic F, Lesson 20
3.CE.2d	Demonstrate fluency with multiplication facts through 10×10 by applying reasoning strategies (e.g., doubling, add-a-group, subtract-a-group, near squares, and inverse relationships).	Mission 1, Topic D, Lesson 11-13 Mission 1, Topic E, Lesson 14, 16 Mission 3, Topic B, Lesson 4-5 Mission 3, Topic C, Lesson 10 Mission 3, Topic D, Lesson 12-13 Mission 3, Topic E, Lesson 16-17 Mission 3, Topic F, Lesson 19

VIRGINIA'S MATHEMATICS STANDARDS OF LEARNING		ZEARN MATH
3rd Grade Standards		Lessons
3.CE.2e	Represent, solve, and justify solutions to single-step contextual problems that involve multiplication and division of whole numbers through 10×10 .	Mission 1, Topic B, Lesson 4-6 Mission 1, Topic C, Lesson 8-9 Mission 1, Topic D, Lesson 11-13 Mission 1, Topic E, Lesson 14-17 Mission 1, Topic F, Lesson 18 Mission 3, Topic A, Lesson 2-3 Mission 3, Topic B, Lesson 7 Mission 3, Topic C, Lesson 11 Mission 3, Topic D, Lesson 15
3.CE.2f	Recall with automaticity the multiplication facts through 10×10 and the corresponding division facts.	Mission 1, Topic D, Lesson 11-13 Mission 1, Topic E, Lesson 14, 16 Mission 3, Topic B, Lesson 4-5 Mission 3, Topic C, Lesson 10 Mission 3, Topic D, Lesson 12-13 Mission 3, Topic E, Lesson 16-17 Mission 3, Topic F, Lesson 19
3.CE.2g	Create an equation to represent the mathematical relationship between equivalent expressions using multiplication and/or division facts through 10×10 (e.g., $4 \times 3 = 14 - 2$, $35 \div 5 = 1 \times 7$).	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
Measurement and Geometry		
3.MG.1a	Justify whether an estimate or an exact measurement is needed for a contextual situation and choose an appropriate unit.	Mission 3, Topic C, Lesson 12
3.MG.1b	Estimate and measure: i. length of an object to the nearest U.S. Customary unit (1/2 inch, inch, foot, yard) and metric unit (centimeter, meter); ii. weight/mass of an object to the nearest U.S. Customary unit (pound) and metric unit (kilogram); and iii. liquid volume to the nearest U.S. Customary unit (cup, pint, quart, gallon) and metric unit (liter).	Mission 2, Topic B, Lesson 7-11 Mission 2, Topic C, Lesson 12 Mission 2, Topic D, Lesson 15-17 Mission 2, Topic E, Lesson 18-21 Mission 3, Topic B, Lesson 6
3.MG.1c	Compare estimates of length, weight/mass, or liquid volume with the actual measurements.	Mission 3, Topic B, Lesson 7-8, 10 Mission 3, Topic E, Lesson 20-21

VIRGINIA'S MATHEMATICS STANDARDS OF LEARNING		ZEARN MATH
3rd Grade Standards		Lessons
3.MG.2a	Solve problems, including those in context, involving area: i. describe and give examples of area as a measurement in contextual situations; and ii. estimate and determine the area of a given surface by counting the number of square units, describe the measurement (using the number and unit) and justify the measurement.	Mission 4, Topic A, Lesson 1-4 Mission 4, Topic B, Lesson 5-8 Mission 4, Topic C, Lesson 9-11 Mission 4, Topic D, Lesson 12-16 Mission 7, Topic E, Lesson 28-30
3.MG.2b	Solve problems, including those in context, involving perimeter: i. describe and give examples of perimeter as a measurement in contextual situations; ii. estimate and measure the distance around a polygon (with no more than six sides) to determine the perimeter and justify the measurement; and iii. given the lengths of all sides of a polygon (with no more than six sides), determine its perimeter and justify the measurement.	Mission 7, Topic C, Lesson 10-17 Mission 7, Topic D, Lesson 18-22 Mission 7, Topic E, Lesson 23-30
3.MG.3a	Tell and write time to the nearest minute, using analog and digital clocks.	Mission 2, Topic A, Lesson 1-5 Mission 2, Topic D, Lesson 15-17 Mission 2, Topic E, Lesson 18, 21
3.MG.3b	Match a written time (e.g., 4:38, 7:09, 12:51) to the time shown on analog and digital clocks to the nearest minute.	Mission 2, Topic A, Lesson 1-5 Mission 2, Topic D, Lesson 15-17 Mission 2, Topic E, Lesson 18, 21
3.MG.3c	Solve single-step contextual problems involving elapsed time in one-hour increments, within a 12-hour period (within a.m. or within p.m.) when given: i. the starting time and the ending time, determine the amount of time that has elapsed; ii. the starting time and amount of elapsed time in one-hour increments, determine the ending time; or iii. the ending time and the amount of elapsed time in one-hour increments, determine the starting time.	Mission 2, Topic A, Lesson 1-5 Mission 2, Topic D, Lesson 15-17 Mission 2, Topic E, Lesson 18, 21

VIRGINIA'S MATHEMATICS STANDARDS OF LEARNING		ZEARN MATH
3rd Grade Standards		Lessons
3.MG.4a	Describe a polygon as a closed plane figure composed of at least three line segments that do not cross.	Mission 7, Topic B, Lesson 4-9
3.MG.4b	Classify figures as polygons or not polygons and justify reasoning.	Mission 7, Topic B, Lesson 4-9
3.MG.4c	Identify and describe triangles, quadrilaterals, pentagons, hexagons, and octagons in various orientations, with and without contexts.	Mission 7, Topic B, Lesson 4-9
3.MG.4d	Identify and name examples of polygons (triangles, quadrilaterals, pentagons, hexagons, octagons) in the environment.	Mission 7, Topic B, Lesson 4-9
3.MG.4e	Classify and compare polygons (triangles, quadrilaterals, pentagons, hexagons, octagons).	Mission 7, Topic B, Lesson 4-9
3.MG.4f	Combine no more than three polygons, where each has three or four sides, and name the resulting polygon (triangles, quadrilaterals, pentagons, hexagons, octagons).	Mission 7, Topic B, Lesson 4-9
3.MG.4g	Subdivide a three-sided or four-sided polygon into no more than three parts and name the resulting polygons.	Mission 7, Topic B, Lesson 4-9
Probability and Statistics		
3.PS.1a	Formulate questions that require the collection or acquisition of data.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
3.PS.1b	Determine the data needed to answer a formulated question and collect or acquire existing data (limited to 30 or fewer data points for no more than eight categories) using various methods (e.g., polls, observations, tallies).	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
3.PS.1c	Organize and represent a data set using pictographs that include an appropriate title, labeled axes, and key. Each pictograph symbol should represent 1, 2, 5 or 10 data points.	Mission 6, Topic A, Lesson 1-4

VIRGINIA'S MATHEMATICS STANDARDS OF LEARNING		ZEARN MATH
3rd Grade Standards		Lessons
3.PS.1d	Organize and represent a data set using bar graphs with a title and labeled axes, with and without the use of technology tools. Determine and use an appropriate scale (increments limited to multiples of 1, 2, 5 or 10)	Mission 6, Topic A, Lesson 1-4 <i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
3.PS.1e	Analyze data represented in pictographs and bar graphs, and communicate results orally and in writing: i. describe the categories of data and the data as a whole (e.g., data were collected on preferred ways to cook or prepare eggs - scrambled, fried, hard boiled, and egg salad); ii. identify parts of the data that have special characteristics, including categories with the greatest, the least, or the same (e.g., most students prefer scrambled eggs); iii. make inferences about data represented in pictographs and bar graphs; iv. use characteristics of the data to draw conclusions about the data and make predictions based on the data (e.g., it is unlikely that a third grader would like hard boiled eggs); and v. solve one- and two-step addition and subtraction problems using data from pictographs and bar graphs.	Mission 6, Topic A, Lesson 1-4
Patterns, Functions, and Algebra		
3.PFA.1a	Identify and describe increasing and decreasing patterns using various representations (e.g., objects, pictures, numbers, number lines).	Mission 3, Topic A, Lesson 1 Mission 3, Topic D, Lesson 13-14 Mission 3, Topic E, Lesson 16-17 Mission 3, Topic F, Lesson 19
3.PFA.1b	Analyze an increasing or decreasing pattern and generalize the change to extend the pattern or identify missing terms using various representations.	Mission 3, Topic A, Lesson 1 Mission 3, Topic D, Lesson 13-14 Mission 3, Topic E, Lesson 16-17 Mission 3, Topic F, Lesson 19
3.PFA.1c	Solve contextual problems that involve identifying, describing, and extending patterns.	Mission 3, Topic A, Lesson 1 Mission 3, Topic D, Lesson 13-14 Mission 3, Topic E, Lesson 16-17 Mission 3, Topic F, Lesson 19

VIRGINIA'S MATHEMATICS STANDARDS OF LEARNING		ZEARN MATH
3rd Grade Standards		Lessons
3.PFA.1d	Create increasing and decreasing patterns using objects, pictures, numbers, and number lines.	Mission 3, Topic A, Lesson 1 Mission 3, Topic D, Lesson 13-14 Mission 3, Topic E, Lesson 16-17 Mission 3, Topic F, Lesson 19
3.PFA.1e	Investigate and explain the connection between two different representations of the same increasing or decreasing pattern.	Mission 3, Topic A, Lesson 1 Mission 3, Topic D, Lesson 13-14 Mission 3, Topic E, Lesson 16-17 Mission 3, Topic F, Lesson 19

4th Grade

VIRGINIA'S MATHEMATICS STANDARDS OF LEARNING		ZEARN MATH
4th Grade Standards		Lessons
Number and Number Sense		
4.NS.1a	Read nine-digit whole numbers, presented in standard form, and represent the same number in written form.	Mission 1, Topic B, Lesson 4
4.NS.1b	Write nine-digit whole numbers in standard form when the numbers are presented orally or in written form.	Mission 1, Topic A, Lesson 3 Mission 1, Topic B, Lesson 4
4.NS.1c	Apply patterns within the base 10 system to determine and communicate, orally and in written form, the place and value of each digit in a nine-digit whole number (e.g., in 568,165,724, the 8 represents 8 millions and its value is 8,000,000).	Mission 1, Topic B, Lesson 6
4.NS.2a	Compare two whole numbers up to seven digits each, using words (greater than, less than, equal to, not equal to) and/or using symbols ($>$, $<$, $=$, $\neq$).	Mission 1, Topic B, Lesson 5
4.NS.2b	Order up to four whole numbers up to seven digits each, from least to greatest or greatest to least.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
4.NS.3a	Compare and order no more than four fractions (proper or improper), and/or mixed numbers, with like denominators by comparing the number of parts (numerators) using fractions with denominators of 12 or less (e.g., $1/5 < 3/5$). Justify comparisons orally, in writing, or with a model.	Mission 5, Topic E, Lesson 26-27
4.NS.3b	Compare and order no more than four fractions (proper or improper), and/or mixed numbers, with like numerators and unlike denominators by comparing the size of the parts using fractions with denominators of 12 or less (e.g., $3/8 < 3/5$). Justify comparisons orally, in writing, or with a model.	Mission 5, Topic E, Lesson 26-27
4.NS.3c	Use benchmarks (e.g., 0, $1/2$, or 1) to compare and order no more than four fractions (proper or	Mission 5, Topic C, Lesson 12-13

VIRGINIA'S MATHEMATICS STANDARDS OF LEARNING		ZEARN MATH
4th Grade Standards		Lessons
	improper), and/or mixed numbers, with like and unlike denominators of 12 or less. Justify comparisons orally, in writing, or with a model.	
4.NS.3d	Compare two fractions (proper or improper) and/or mixed numbers using fractions with denominators of 12 or less, using the symbols $>$, $<$, and $=$ (e.g., $\frac{2}{3} > \frac{1}{7}$). Justify comparisons orally, in writing, or with a model.	Mission 5, Topic C, Lesson 12-15 Mission 5, Topic E, Lesson 26-27
4.NS.3e	Represent equivalent fractions with denominators of 12 or less, with and without models.	Mission 5, Topic B, Lesson 7-11 Mission 5, Topic C, Lesson 14-15 Mission 5, Topic D, Lesson 21 Mission 5, Topic E, Lesson 27 Mission 6, Topic B, Lesson 5 Mission 6, Topic D, Lesson 12
4.NS.3f	Compose and decompose fractions (proper and improper) and/or mixed numbers with denominators of 12 or less, in multiple ways, with and without models.	Mission 5, Topic A, Lesson 1-6 Mission 5, Topic D, Lesson 16-18 Mission 5, Topic E, Lesson 22-25 Mission 5, Topic F, Lesson 30-32
4.NS.3g	Represent the division of two whole numbers as a fraction given a contextual situation and a model (e.g., $\frac{3}{5}$ means the same as 3 divided by 5 or $\frac{3}{5}$ represents the amount of muffin each of five children will receive when sharing three muffins equally).	Grade 5, Mission 4, Topic B, Lesson 2-5
4.NS.4a	Investigate and describe the ten-to-one place value relationship for decimals through thousandths, using concrete models (e.g., place value mats/charts, decimal squares, base 10 blocks).	Mission 6, Topic A, Lesson 1, 3 Mission 6, Topic B, Lesson 4-5
4.NS.4b	Represent and identify decimals expressed through thousandths, using concrete, pictorial, and numerical representations.	Mission 6, Topic A, Lesson 1-3 Mission 6, Topic B, Lesson 4-8
4.NS.4c	Read and write decimals expressed through thousandths, using concrete, pictorial, and numerical representations.	Mission 6, Topic A, Lesson 1-3 Mission 6, Topic B, Lesson 4-8
4.NS.4d	Identify and communicate, both orally and in written form, the place and value of each digit in a decimal	Mission 6, Topic A, Lesson 1-3 Mission 6, Topic B, Lesson 4-8

VIRGINIA'S MATHEMATICS STANDARDS OF LEARNING		ZEARN MATH
4th Grade Standards		Lessons
	through thousandths (e.g., given 0.385, the 8 is in the hundredths place and has a value of 0.08).	
4.NS.4e	Compare using symbols ($<$, $>$, $=$) and/or words (greater than, less than, equal to) and order (least to greatest and greatest to least), a set of no more than four decimals expressed through thousandths, using multiple strategies (e.g., benchmarks, place value, number lines). Justify comparisons with a model, orally, and in writing.	Mission 6, Topic C, Lesson 9-11
4.NS.5a	Represent fractions (proper or improper) and/or mixed numbers as decimals through hundredths, using multiple representations, limited to halves, fourths, fifths, tenths, and hundredths.	Mission 6, Topic B, Lesson 4-8
4.NS.5b	Identify and model equivalent relationships between fractions (proper or improper) and/or mixed numbers and decimals, using halves, fourths, fifths, tenths, and hundredths.	Mission 6, Topic A, Lesson 1-3 Mission 6, Topic B, Lesson 4-8
4.NS.5c	Write the decimal and fraction equivalent for a given model (e.g., $1/4 = 0.25$ or $0.25 = 1/4$; $1.25 = 5/4$ or $11/4$; $1.02 = 102/100$ or $1 \frac{2}{100}$).*	Mission 6, Topic A, Lesson 1-3 Mission 6, Topic B, Lesson 4-8
Computation and Estimation		
4.CE.1a	Determine and justify whether an estimate or an exact answer is appropriate when solving contextual problems involving addition and subtraction with whole numbers. Refine estimates by adjusting the final amount, using terms such as closer to, between, and a little more than	Mission 1, Topic C, Lesson 10 Mission 1, Topic D, Lesson 12 Mission 1, Topic E, Lesson 16
4.CE.1b	Apply strategies (e.g., rounding to the nearest 100 or 1,000, using compatible numbers, other number relationships) to estimate a solution for single-step or multi step addition or subtraction problems with whole numbers, where addends or minuends do not exceed 10,000.	Mission 1, Topic C, Lesson 8,10 Mission 1, Topic D, Lesson 12 Mission 1, Topic E, Lesson 16
4.CE.1c	Apply strategies (e.g., place value, properties of addition, other number relationships) and algorithms,	Mission 1, Topic D, Lesson 11-13 Mission 1, Topic E, Lesson 14-16

VIRGINIA'S MATHEMATICS STANDARDS OF LEARNING		ZEARN MATH
4th Grade Standards		Lessons
	including the standard algorithm, to determine the sum or difference of two whole numbers, where addends and minuends do not exceed 10,000.	Mission 1, Topic F, Lesson 17-19
4.CE.1d	Estimate, represent, solve, and justify solutions to single-step and multistep contextual problems involving addition and subtraction with whole numbers where addends and minuends do not exceed 1,000,000.	Mission 1, Topic C, Lesson 10 Mission 1, Topic D, Lesson 12 Mission 1, Topic E, Lesson 16 Mission 1, Topic F, Lesson 17-19
4.CE.2a	Determine and justify whether an estimate or an exact answer is appropriate when solving contextual problems involving multiplication and division of whole numbers. Refine estimates by adjusting the final amount, using terms such as closer to, between, and a little more than.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
4.CE.2b	Recall with automaticity the multiplication facts through 12×12 and the corresponding division facts.	<i>Zearn Math includes Number Gym activities in Independent Digital Lessons that target 4.CE.2b; however, the Number Gym activities only go up to multiplication facts through 10×10. Teachers will need to supplement instruction to ensure students master the full intent of 4.CE.2b.</i>
4.CE.2c	Create an equation using addition, subtraction, multiplication, and division to represent the relationship between equivalent mathematical expressions (e.g., $4 \times 3 = 2 \times 6$; $10 + 8 = 36 \div 2$; $12 \times 4 = 60 - 12$).	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
4.CE.2d	Identify and use the appropriate symbol to distinguish between expressions that are equal and expressions that are not equal, using addition, subtraction, multiplication, and division (e.g., $4 \times 12 = 8 \times 6$ and $64 \div 8 \neq 8 \times 8$).	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
4.CE.2e	Determine all factor pairs for a whole number 1 to 100, using concrete, pictorial, and numerical representations	Mission 3, Topic F, Lesson 22-25

VIRGINIA'S MATHEMATICS STANDARDS OF LEARNING		ZEARN MATH
4th Grade Standards		Lessons
4.CE.2f	Determine common factors and the greatest common factor of no more than three numbers.	Mission 3, Topic F, Lesson 22-25
4.CE.2g	Apply strategies (e.g., rounding, place value, properties of multiplication and/or addition) and algorithms, including the standard algorithm, to estimate and determine the product of two whole numbers when given: i. a two-digit factor and a one-digit factor; ii. a three-digit factor and a one-digit factor; or iii. a two-digit factor and a two-digit factor.	Mission 3, Topic B, Lesson 4-6 Mission 3, Topic C, Lesson 7-11 Mission 3, Topic H, Lesson 34-38
4.CE.2h	Estimate, represent, solve, and justify solutions to single-step and multistep contextual problems that involve multiplication with whole numbers.	Mission 3, Topic A, Lesson 2-3 Mission 3, Topic B, Lesson 5 Mission 3, Topic C, Lesson 9, 11 Mission 3, Topic D, Lesson 12-13 Mission 7, Topic A, Lesson 4 Mission 7, Topic C, Lesson 14
4.CE.2i	Apply strategies (e.g., rounding, compatible numbers, place value) and algorithms, including the standard algorithm, to estimate and determine the quotient of two whole numbers, given a one-digit divisor and a two- or three-digit dividend, with and without remainders.	Mission 3, Topic E, Lesson 14-21 Mission 3, Topic G, Lesson 26-33
4.CE.2j	Estimate, represent, solve, and justify solutions to single-step contextual problems involving division with whole numbers.	Mission 3, Topic E, Lesson 14, 19 Mission 3, Topic G, Lesson 26, 28-29, 31-32
4.CE.2k	Interpret the quotient and remainder when solving a contextual problem.	Mission 3, Topic E, Lesson 14, 19 Mission 3, Topic G, Lesson 26, 28-29, 31-32
4.CE.3a	Estimate and determine the sum or difference of two fractions (proper or improper) and/or mixed numbers, having like denominators limited to 2, 3, 4, 5, 6, 8, 10, and 12 (e.g., $\frac{3}{8} + \frac{3}{8}$, $2\frac{1}{5} + \frac{4}{5}$, $\frac{7}{4} - \frac{5}{4}$) and simplify the resulting fraction. Addition and subtraction with fractions may include regrouping.	Mission 5, Topic D, Lesson 16-21 Mission 5, Topic E, Lesson 22 Mission 5, Topic F, Lesson 29-34
4.CE.3b	Estimate, represent, solve, and justify solutions to single-step contextual problems using addition and	Mission 5, Topic D, Lesson 19

VIRGINIA'S MATHEMATICS STANDARDS OF LEARNING		ZEARN MATH
4th Grade Standards		Lessons
	subtraction with fractions (proper or improper) and/or mixed numbers, having like denominators limited to 2, 3, 4, 5, 6, 8, 10, and 12, and simplify the resulting fraction. Addition and subtraction with fractions may include regrouping.	
4.CE.3c	Solve single-step contextual problems involving multiplication of a whole number, limited to 12 or less, and a unit fraction (e.g., $6 \times \frac{1}{3}$, $15 \times \frac{1}{8}$, $2 \times \frac{1}{10}$), with models.	Mission 5, Topic G, Lesson 39-40
4.CE.3d	Apply the inverse property of multiplication in models (e.g., use a visual fraction model to represent $\frac{4}{4}$ or 1 as the product of $4 \times \frac{1}{4}$).	Mission 5, Topic E, Lesson 24-25 Mission 5, Topic G, Lesson 35-38
4.CE.4a	Apply strategies (e.g., rounding to the nearest whole number, using compatible numbers) and algorithms, including the standard algorithm, to estimate and determine the sum or difference of two decimals through the thousandths, with and without models, in which: i. decimals do not exceed the thousandths; and ii. addends, subtrahends, and minuends are limited to four digits.	Mission 6, Topic D, Lesson 14 <i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
4.CE.4b	Estimate, represent, solve, and justify solutions to single-step and multistep contextual problems using addition and subtraction of decimals through the thousandths.	Mission 6, Topic D, Lesson 14 <i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
Measurement and Geometry		
4.MG.1a	Determine an appropriate unit of measure to use when measuring: i. length in both U.S. Customary (inch, foot, yard, mile) and metric units (millimeter, centimeter, meter); ii. weight/mass in both U.S. Customary (ounce, pound) and metric units (gram, kilogram); and iii. liquid volume in both U.S. Customary (cup, pint, quart, gallon) and metric units (milliliter, liter).	Mission 7, Topic A, Lesson 1-2 <i>It is recommended that teachers use supplemental materials to fully address this standard.</i>

VIRGINIA'S MATHEMATICS STANDARDS OF LEARNING		ZEARN MATH
4th Grade Standards		Lessons
4.MG.1b	Estimate and measure: i. length of an object to the nearest U.S. Customary unit (1/2 inch, 1/4 inch, 1/8 inch, foot, yard) and nearest metric unit (millimeter, centimeter, or meter); ii. weight/mass of an object to the nearest U.S. Customary unit (ounce, pound) and nearest metric unit (gram, kilogram); and iii. liquid volume to the nearest U.S. Customary unit (cup, pint, quart, gallon) and nearest metric unit (milliliter, liter).	Mission 7, Topic A, Lesson 1-2 <i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
4.MG.1c	Compare estimates of length, weight/mass, or liquid volume with the actual measurements.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
4.MG.1d	Given the equivalent measure of one unit, solve problems, including those in context, by determining the equivalent measures within the U.S. Customary system for: i. length (inches and feet, feet and yards, inches and yards); ii. weight/mass (ounces and pounds); and iii. liquid volume (cups, pints, quarts, and gallons).	Mission 7, Topic A, Lesson 1-2, 4-5 Mission 7, Topic B, Lesson 6-8, 10-11 Mission 7, Topic C, Lesson 12-14
4.MG.2a	Solve single-step and multistep contextual problems involving elapsed time in hours and minutes, within a 12-hour period (within a.m., within p.m., and across a.m. and p.m.) when given: i. the starting time and the ending time, determine the amount of time that has elapsed in hours and minutes; ii. the starting time and amount of elapsed time in hours and minutes, determine the ending time; or iii. the ending time and the amount of elapsed time in hours and minutes, determine the starting time.	Mission 7, Topic A, Lesson 3 <i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
4.MG.3a	Use concrete materials and pictorial models to develop a formula for the area and perimeter of a rectangle (including a square).	Mission 3, Topic A, Lesson 1

VIRGINIA'S MATHEMATICS STANDARDS OF LEARNING		ZEARN MATH
4th Grade Standards		Lessons
4.MG.3b	Determine the area and perimeter of a rectangle when given the measure of two adjacent sides (in whole number units), with and without models.	Mission 3, Topic A, Lesson 1-3
4.MG.3c	Determine the area and perimeter of a square when given the measure of one side (in whole number units), with and without models.	Mission 3, Topic A, Lesson 2
4.MG.3d	Use concrete materials and pictorial models to explore the relationship between area and perimeter of rectangles.	Mission 3, Topic A, Lesson 2-3
4.MG.3e	Identify and represent rectangles with the same perimeter and different areas or with the same area and different perimeters.	Mission 3, Topic A, Lesson 1
4.MG.3f	Solve contextual problems involving area and perimeter of rectangles and squares.	Mission 3, Topic A, Lesson 2-3
4.MG.4a	Identify and describe points, lines, line segments, rays, and angles, including endpoints and vertices.	Mission 4, Topic A, Lesson 1-4
4.MG.4b	Describe endpoints and vertices in relation to lines, line segments, rays, and angles.	Mission 4, Topic A, Lesson 1-4
4.MG.4c	Draw representations of points, line segments, rays, angles, and lines, using a ruler or straightedge.	Mission 4, Topic A, Lesson 1-4
4.MG.4d	Identify parallel, perpendicular, and intersecting lines and line segments in plane and solid figures, including those in context.	Mission 4, Topic A, Lesson 3-4
4.MG.4e	Use symbolic notation to name points, lines, line segments, rays, angles, and to describe parallel and perpendicular lines.	Mission 4, Topic A, Lesson 1-4
4.MG.5a	Develop definitions for parallelograms, rectangles, squares, rhombi, and trapezoids through the exploration of properties and attributes.	Mission 4, Topic D, Lesson 15-16
4.MG.5b	Identify and describe points, line segments, angles, and vertices in quadrilaterals.	Mission 4, Topic D, Lesson 15-16

VIRGINIA'S MATHEMATICS STANDARDS OF LEARNING		ZEARN MATH
4th Grade Standards		Lessons
4.MG.5c	Identify and describe parallel, intersecting, perpendicular, and congruent sides in quadrilaterals.	Mission 4, Topic D, Lesson 15-16
4.MG.5d	Compare, contrast, and classify quadrilaterals (parallelograms, rectangles, squares, rhombi, and/or trapezoids) based on the following properties and attributes: i. parallel sides; ii. perpendicular sides; iii. congruence of sides; and iv. number of right angles.	Mission 4, Topic D, Lesson 15-16
4.MG.5e	Denote properties of quadrilaterals and identify parallel sides, congruent sides, and right angles by using geometric markings.	Mission 4, Topic D, Lesson 15-16
4.MG.5f	Use symbolic notation to name line segments and angles in quadrilaterals.	Mission 4, Topic D, Lesson 15-16
4.MG.6a	Identify concrete models and pictorial representations of solid figures (cube, rectangular prism, square pyramid, sphere, cone, and cylinder).	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
4.MG.6b	Identify and describe solid figures (cube, rectangular prism, square pyramid, and sphere) according to their characteristics (number of angles, vertices, edges, and by the number and shape of faces).	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
4.MG.6c	Compare and contrast plane and solid figures (limited to circles, squares, triangles, rectangles, spheres, cubes, square pyramids, and rectangular prisms) according to their characteristics (number of sides, angles, vertices, edges, and the number and shape of faces).	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
Probability and Statistics		
4.PS.1a	Formulate questions that require the collection or acquisition of data.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>

VIRGINIA'S MATHEMATICS STANDARDS OF LEARNING		ZEARN MATH
4th Grade Standards		Lessons
4.PS.1b	Determine the data needed to answer a formulated question and collect or acquire existing data (limited to 10 or fewer data points) using various methods (e.g., observations, measurements, experiments).	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
4.PS.1c	Organize and represent a data set using line graphs with a title and labeled axes with whole number increments, with and without the use of technology tools.	Mission 5, Topic E, Lesson 28 Mission 5, Topic G, Lesson 40
4.PS.1d	Analyze data represented in line graphs and communicate results orally and in writing: i. describe the characteristics of the data represented in a line graph and the data as a whole (e.g., the time period when the temperature increased the most); ii. identify parts of the data that have special characteristics and explain the meaning of the greatest, the least, or the same (e.g., the highest temperature shows the warmest day); iii. make inferences about data represented in line graphs; iv. draw conclusions about the data and make predictions based on the data to answer questions; and v. solve single-step and multi step addition and subtraction problems using data from line graphs.	Mission 5, Topic E, Lesson 28 Mission 5, Topic G, Lesson 40
4.PS.2a	Describe probability as the degree of likelihood of an outcome occurring using terms such as impossible, unlikely, equally likely, likely, and certain.	Grade 7, Mission 8, Topic A, Lesson 2-3, 5 <i>Zearn includes lessons on probability in Grade 7, however it is recommended that teachers use supplemental materials to fully address this standard.</i>
4.PS.2b	Model and determine all possible outcomes of a given simple event where there are no more than 24 possible outcomes, using a variety of manipulatives (e.g., coins, two-sided counters, number cubes, spinners).	Grade 7, Mission 8, Topic A, Lesson 3-6 <i>Zearn includes lessons on probability in Grade 7, however it is recommended that teachers</i>

VIRGINIA'S MATHEMATICS STANDARDS OF LEARNING		ZEARN MATH
4th Grade Standards		Lessons
		<i>use supplemental materials to fully address this standard.</i>
4.PS.2c	Write the probability of a given simple event as a fraction between 0 and 1, where there are no more than 24 possible outcomes.	Grade 7, Mission 8, Topic A, Lesson 3-6 <i>Zearn includes lessons on probability in Grade 7, however it is recommended that teachers use supplemental materials to fully address this standard.</i>
4.PS.2d	Determine the likelihood of an event occurring and relate it to its whole number or fractional representation (e.g., impossible or zero; equally likely; certain or one).	<i>Zearn includes lessons on probability in Grade 7, however it is recommended that teachers use supplemental materials to fully address this standard.</i>
4.PS.2e	Create a model or contextual problem to represent a given probability.	Grade 7, Mission 8, Topic A, Lesson 2-3, 5 <i>Zearn includes lessons on probability in Grade 7, however it is recommended that teachers use supplemental materials to fully address this standard.</i>
Patterns, Functions, and Algebra		
4.PFA.1a	Identify, describe, extend, and create increasing and decreasing patterns using various representations (e.g., objects, pictures, numbers, number lines, input/output tables, and function machines).	Mission 5, Topic H, Lesson 41
4.PFA.1b	Analyze an increasing or decreasing single-operation numerical pattern found in lists, input/output tables, or function machines and generalize the change to identify the rule, extend the pattern, or identify missing terms.	Mission 5, Topic H, Lesson 41
4.PFA.1c	Given a rule, create increasing and decreasing patterns using numbers and input/output tables (including function machines).	Mission 5, Topic H, Lesson 41

VIRGINIA'S MATHEMATICS STANDARDS OF LEARNING		ZEARN MATH
4th Grade Standards		Lessons
4.PFA.1d	Solve contextual problems that involve identifying, describing, and extending increasing and decreasing patterns using single-operation input and output rules.	Mission 5, Topic H, Lesson 41

5th Grade

VIRGINIA'S MATHEMATICS STANDARDS OF LEARNING		ZEARN MATH
5th Grade Standards		Lessons
Number and Number Sense		
5.NS.1a	Use concrete and pictorial models to represent fractions with denominators that are thirds, eighths, and factors of 100 in their equivalent decimal form.	Grade 4, Mission 6, Topic A, Lesson 1-3 Grade 4, Mission 6, Topic B, Lesson 4-6 Grade 4, Mission 6, Topic C, Lesson 11 Grade 4, Mission 6, Topic D, Lesson 12-14 Grade 4, Mission 6, Topic E, Lesson 15
5.NS.1b	Use concrete and pictorial models to represent decimals in their equivalent fraction form (thirds, eighths, and factors of 100).	Grade 4, Mission 6, Topic A, Lesson 1-3 Grade 4, Mission 6, Topic B, Lesson 4-6 Grade 4, Mission 6, Topic C, Lesson 11 Grade 4, Mission 6, Topic D, Lesson 12-14 Grade 4, Mission 6, Topic E, Lesson 15
5.NS.1c	Identify equivalent relationships between decimals and fractions with denominators that are thirds, eighths, and factors of 100 in their equivalent decimal form, with and without models.	Grade 4, Mission 6, Topic A, Lesson 1-3 Grade 4, Mission 6, Topic B, Lesson 4-6 Grade 4, Mission 6, Topic C, Lesson 11 Grade 4, Mission 6, Topic D, Lesson 12-14 Grade 4, Mission 6, Topic E, Lesson 15
5.NS.1d	Compare (using symbols $<$, $>$, $=$) and order (least to greatest and greatest to least) a set of no more than four decimals and fractions (proper, improper) and/or mixed numbers using multiple strategies (e.g.,	Mission 1, Topic B, Lesson 6

VIRGINIA'S MATHEMATICS STANDARDS OF LEARNING		ZEARN MATH
5th Grade Standards		Lessons
	benchmarks, place value, number lines). Justify solutions orally, in writing, or with a model.	
5.NS.2a	Given a whole number up to 100, create a concrete or pictorial representation to demonstrate whether the number is prime or composite, and justify reasoning.	Grade 4, Mission 3, Topic F, Lesson 22-25
5.NS.2b	Classify, compare, and contrast whole numbers up to 100 using the characteristics prime and composite.	Grade 4, Mission 3, Topic F, Lesson 22-25
5.NS.2c	Determine the prime factorization for a whole number up to 100.	Grade 4, Mission 3, Topic F, Lesson 22-25
Computation and Estimation		
5.CE.1a	Estimate the sum, difference, product, and quotient of whole numbers in contextual problems.	Grade 4, Mission 3, Topic D, Lesson 12-13 Grade 4, Mission 3, Topic E, Lesson 14, 19 Grade 4, Mission 3, Topic G, Lesson 31-32 Mission 2, Topic A, Lesson 2
5.CE.1b	Represent, solve, and justify solutions to single-step and multistep contextual problems by applying strategies (e.g., estimation, properties of addition and multiplication) and algorithms, including the standard algorithm, involving addition, subtraction, multiplication, and division of whole numbers, with and without remainders, in which: i. sums, differences, and products do not exceed five digits; ii. factors do not exceed two digits by three digits; iii. divisors do not exceed two digits; or iv. dividends do not exceed four digits.	Grade 4, Mission 3, Topic D, Lesson 12-13 Grade 4, Mission 3, Topic E, Lesson 14, 19 Grade 4, Mission 3, Topic G, Lesson 31-32
5.CE.1c	Interpret the quotient and remainder when solving a contextual problem.	Grade 4, Mission 3, Topic D, Lesson 12-13 Grade 4, Mission 3, Topic E, Lesson 14, 19 Grade 4, Mission 3, Topic G, Lesson 31-32

VIRGINIA'S MATHEMATICS STANDARDS OF LEARNING		ZEARN MATH
5th Grade Standards		Lessons
5.CE.2a	Determine the least common multiple of two numbers to find the least common denominator for two fractions.	Grade 6, Mission 7, Topic D, Lesson 16-18
5.CE.2b	Estimate and determine the sum or difference of two fractions (proper or improper) and/or mixed numbers, having like and unlike denominators limited to 2, 3, 4, 5, 6, 8, 10, and 12 (e.g., $5/8 + 1/4$, $4/5 - 2/3$, $3\ 3/4 + 2\ 5/12$), and simplify the resulting fraction.	Mission 3, Topic A, Lesson 1-2 Mission 3, Topic B, Lesson 3-6 Mission 3, Topic C, Lesson 8-12 Mission 3, Topic D, Lesson 14
5.CE.2c	Estimate and solve single-step and multistep contextual problems involving addition and subtraction with fractions (proper or improper) and/or mixed numbers having like and unlike denominators, with and without models. Denominators should be limited to 2, 3, 4, 5, 6, 8, 10, and 12. Answers should be expressed in simplest form.	Mission 3, Topic B, Lesson 3-7 Mission 3, Topic C, Lesson 8-12 Mission 3, Topic D, Lesson 13-15 Mission 4, Topic D, Lesson 10-12
5.CE.2d	Solve single-step contextual problems involving multiplication of a whole number, limited to 12 or less, and a proper fraction (e.g., $9 \times 2/3$, $8 \times 3/4$), with models. The denominator will be a factor of the whole number and answers should be expressed in simplest form.	Mission 3, Topic B, Lesson 3-7 Mission 3, Topic C, Lesson 8-12 Mission 3, Topic D, Lesson 13-15 Mission 4, Topic D, Lesson 10-12
5.CE.3a	Apply estimation strategies (e.g., rounding to the nearest whole number, tenth or hundredth; compatible numbers, place value) to determine a reasonable solution for single-step and multistep contextual problems involving addition, subtraction, and multiplication of decimals, and single-step contextual problems involving division of decimals.	Mission 1, Topic F, Lesson 6 Mission 2, Topic D, Lesson 13-14 Mission 2, Topic G, Lesson 27 Mission 2, Topic H, Lesson 28-29
5.CE.3b	Estimate and determine the product of two numbers using strategies and algorithms, including the standard algorithm, when given: i. a two-digit factor and a one-digit factor (e.g., 2.3×4; 0.08×0.9; $.16 \times 5$); ii. a three-digit factor and a one-digit factor (e.g., 0.156×4, 3.28×7, 8.09×0.2); and iii. a two-digit factor and a two-digit factor (e.g., 0.85×3.7, 14×1.6, 9.2×3.5). (Products will not exceed the thousandths place, and	Mission 1, Topic E, Lesson 11-12 Mission 2, Topic C, Lesson 10-12 Mission 2, Topic D, Lesson 13-14 Mission 4, Topic E, Lesson 17-18

VIRGINIA'S MATHEMATICS STANDARDS OF LEARNING		ZEARN MATH
5th Grade Standards		Lessons
	leading zeroes will not be considered when counting digits.)	
5.CE.3c	Estimate and determine the quotient of two numbers using strategies and algorithms, including the standard algorithm, in which: i. quotients do not exceed four digits with or without a decimal point; ii. quotients may include whole numbers, tenths, hundredths, or thousandths; iii. divisors are limited to a single digit whole number or a decimal expressed as tenths; and iv. no more than one additional zero will need to be annexed.	Mission 1, Topic F, Lesson 13-16 Mission 2, Topic E, Lesson 16-18 Mission 2, Topic F, Lesson 19-23 Mission 2, Topic G, Lesson 24-27 Mission 2, Topic H, Lesson 28-29 Mission 4, Topic G, Lesson 29-31
5.CE.3d	Solve single-step and multistep contextual problems involving addition, subtraction, and multiplication of decimals by applying strategies (e.g., estimation, modeling) and algorithms, including the standard algorithm.	Mission 1, Topic F, Lesson 16 Mission 2, Topic D, Lesson 13-14 Mission 2, Topic G, Lesson 27 Mission 2, Topic H, Lesson 28-29
5.CE.3e	Solve single-step contextual problems involving division with decimals by applying strategies (e.g., estimation, modeling) and algorithms, including the standard algorithm.	Mission 1, Topic F, Lesson 16 Mission 2, Topic G, Lesson 27 Mission 2, Topic H, Lesson 28-29
5.CE.4a	Use order of operations to simplify numerical expressions with whole numbers, limited to addition, subtraction, multiplication, and division in which: i. expressions may contain no more than one set of parentheses; ii. simplification will be limited to five whole numbers and four operations in any combination of addition, subtraction, multiplication, or division; iii. whole numbers will be limited to two digits or less; and iv. expressions should not include braces, brackets, or fraction bars.	Mission 2, Topic A, Lesson 1 Mission 2, Topic B, Lesson 3-5 Mission 4, Topic D, Lesson 10 Mission 4, Topic H, Lesson 32
5.CE.4b	Given a whole number numerical expression involving more than one operation, describe which operation is completed first, which is second, and which is third.	Mission 2, Topic A, Lesson 1 Mission 2, Topic B, Lesson 3-5 Mission 4, Topic D, Lesson 10

VIRGINIA'S MATHEMATICS STANDARDS OF LEARNING		ZEARN MATH
5th Grade Standards		Lessons
		Mission 4, Topic H, Lesson 32
Measurement and Geometry		
5.MG.1a	Determine the most appropriate unit of measure to use in a contextual problem that involves metric units: i. length (millimeters, centimeters, meters, and kilometers); ii. mass (grams and kilograms); and iii. liquid volume (milliliters and liters).	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
5.MG.1b	Estimate and measure to solve contextual problems that involve metric units: i. length (millimeters, centimeters, and meters); ii. mass (grams and kilograms); and iii. liquid volume (milliliters and liters).	Mission 1, Topic A, Lesson 4 Mission 2, Topic D, Lesson 13-15 Mission 4, Topic C, Lesson 9 Mission 4, Topic E, Lesson 19-20 Mission 5, Topic B, Lesson 5
5.MG.1c	Given the equivalent metric measure of one unit, in a contextual problem, determine the equivalent measurement within the metric system: i. length (millimeters, centimeters, meters, and kilometers); ii. mass (grams and kilograms); and iii. liquid volume (milliliters and liters).	Mission 1, Topic A, Lesson 4 Mission 2, Topic D, Lesson 13-15 Mission 4, Topic C, Lesson 9 Mission 4, Topic E, Lesson 19-20 Mission 5, Topic B, Lesson 5
5.MG.2a	Investigate and develop a formula for determining the area of a right triangle.	Grade 6, Mission 1, Topic A, Lesson 2-3 Grade 6, Mission 1, Topic B, Lesson 4-6 Grade 6, Mission 1, Topic C, Lesson 7-10 Grade 6, Mission 1, Topic D, Lesson 11 Grade 6, Mission 1, Topic G, Lesson 19 Grade 6, Mission 4, Topic D, Lesson 14
5.MG.2b	Estimate and determine the area of a right triangle, with diagrams, when the base and the height are given in whole number units, in metric or U.S. Customary units, and record the solution with the appropriate unit of measure (e.g., 16 square inches).	Grade 6, Mission 1, Topic A, Lesson 2-3 Grade 6, Mission 1, Topic B, Lesson 4-6 Grade 6, Mission 1, Topic C, Lesson 7-10

VIRGINIA'S MATHEMATICS STANDARDS OF LEARNING		ZEARN MATH
5th Grade Standards		Lessons
		Grade 6, Mission 1, Topic D, Lesson 11 Grade 6, Mission 1, Topic G, Lesson 19 Grade 6, Mission 4, Topic D, Lesson 14
5.MG.2c	Describe volume as a measure of capacity and give examples of volume as a measurement in contextual situations.	Mission 5, Topic A, Lesson 1-3 Mission 5, Topic B, Lesson 5
5.MG.2d	Investigate and develop a formula for determining the volume of rectangular prisms using concrete objects.	Mission 5, Topic B, Lesson 4-9
5.MG.2e	Solve problems, including those in context, to estimate and determine the volume of a rectangular prism using concrete objects, diagrams, and formulas when the length, width, and height are given in whole number units. Record the solution with the appropriate unit of measure (e.g., 12 cubic inches).	Mission 5, Topic B, Lesson 4-9
5.MG.2f	Identify whether the application of the concept of perimeter, area, or volume is appropriate for a given situation.	Mission 5, Topic B, Lesson 7 Mission 5, Topic C, Lesson 10-11, 13-15 <i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
5.MG.2g	Solve contextual problems that involve perimeter, area, and volume in standard units of measure.	Mission 5, Topic B, Lesson 7 Mission 5, Topic C, Lesson 10-11, 13-15
5.MG.3a	Classify angles as right, acute, obtuse, or straight and justify reasoning.	Grade 4, Mission 4, Topic A, Lesson 1-4 Grade 4, Mission 4, Topic D, Lesson 13-15
5.MG.3b	Classify triangles as right, acute, or obtuse and equilateral, scalene, or isosceles and justify reasoning.	Grade 4, Mission 4, Topic D, Lesson 13-15

VIRGINIA'S MATHEMATICS STANDARDS OF LEARNING		ZEARN MATH
5th Grade Standards		Lessons
5.MG.3c	Identify congruent sides and right angles using geometric markings to denote properties of triangles.	Grade 4, Mission 4, Topic D, Lesson 14-15
5.MG.3d	Compare and contrast the properties of triangles.	Grade 4, Mission 4, Topic D, Lesson 13-15
5.MG.3e	Identify the appropriate tools (e.g., protractor, straightedge, angle ruler, available technology) to measure and draw angles.	Grade 4, Mission 4, Topic B, Lesson 5-8
5.MG.3f	Measure right, acute, obtuse, and straight angles, using appropriate tools, and identify measures in degrees.	Grade 4, Mission 4, Topic B, Lesson 5-8
5.MG.3g	Use models to prove that the sum of the interior angles of a triangle is 180 degrees and use the relationship to determine an unknown angle measure in a triangle.	Grade 8, Mission 1, Lesson 15-16 <i>Zearn includes lessons on the interior angles of a triangle in Grade 8, however it is recommended that teachers use supplemental materials to fully address this standard.</i>
5.MG.3h	Solve addition and subtraction contextual problems to determine unknown angle measures on a diagram.	Grade 4, Mission 4, Topic C, Lesson 9-11
Probability and Statistics		
5.PS.1a	Formulate questions that require the collection or acquisition of data.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
5.PS.1b	Determine the data needed to answer a formulated question and collect or acquire existing data (limited to 30 or fewer data points) using various methods (e.g., polls, observations, measurements, experiments).	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
5.PS.1c	Organize and represent a data set using a line plot (dot plot) with a title, labeled axes, and a key, with and without the use of technology tools. Line plots (dot plots) may contain whole numbers, fractions, or decimals.	Mission 4, Topic A, Lesson 1 Mission 4, Topic D, Lesson 10

VIRGINIA'S MATHEMATICS STANDARDS OF LEARNING		ZEARN MATH
5th Grade Standards		Lessons
5.PS.1d	Organize and represent numerical data using a stem-and-leaf plot with a title and key, where the stems are listed in ascending order and the leaves are in ascending order, with or without commas between the leaves.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
5.PS.1e	Analyze data represented in line plots (dot plots) and stem-and-leaf plots and communicate results orally and in writing: - describe the characteristics of the data represented in a line plot (dot plot) and stem-and-leaf plot as a whole (e.g., the shape and spread of the data); - make inferences about data represented in line plots (dot plots) and stem-and-leaf plots (e.g., based on a line plot (dot plot) of the number of books students in a bus line have in their backpack, every student will have from two to four books in their backpack); - identify parts of the data that have special characteristics and explain the meaning of the greatest, the least, or the same (e.g., the stem-and-leaf plot shows that the same number of students scored in the 90s as scored in the 70s); - draw conclusions about the data and make predictions based on the data to answer questions; and - solve single-step and multi step addition and subtraction problems using data from line plots (dot plots) and stem-and-leaf plots.	Mission 4, Topic A, Lesson 1 Mission 4, Topic D, Lesson 10
5.PS.2a	Describe mean as fair share.	Grade 6, Mission 8, Topic C, Lesson 9-12
5.PS.2b	Describe and determine the mean of a set of data values representing data from a given context as a measure of center.	Grade 6, Mission 8, Topic C, Lesson 9-12 Grade 6, Mission 8, Topic D, Lesson 14-16 Grade 6, Mission 8, Topic E, Lesson 18

VIRGINIA'S MATHEMATICS STANDARDS OF LEARNING		ZEARN MATH
5th Grade Standards		Lessons
5.PS.2c	Describe and determine the median of a set of data values representing data from a given context as a measure of center.	Grade 6, Mission 8, Topic D, Lesson 13-16 Grade 6, Mission 8, Topic E, Lesson 18
5.PS.2d	Describe and determine the mode of a set of data values representing data from a given context as a measure of center.	Grade 6, Mission 8, Topic A, Lesson 2 Grade 6, Mission 8, Topic B, Lesson 3-7 Grade 6, Mission 8, Topic C, Lesson 9-12 Grade 6, Mission 8, Topic D, Lesson 13-17 Grade 6, Mission 8, Topic E, Lesson 18
5.PS.2e	Describe and determine the range of a set of data values representing data from a given context as a measure of spread.	Grade 6, Mission 8, Topic A, Lesson 2 Grade 6, Mission 8, Topic B, Lesson 3-7 Grade 6, Mission 8, Topic C, Lesson 9-12 Grade 6, Mission 8, Topic D, Lesson 13-17 Grade 6, Mission 8, Topic E, Lesson 18
5.PS.3a	Determine the probability of an outcome by constructing a sample space (with a total of 24 or fewer equally likely possible outcomes), using a tree diagram, list, or chart to represent and determine all possible outcomes.	Grade 7, Mission 8, Topic B, Lesson 8-9
5.PS.3b	Determine the number of possible outcomes by using the Fundamental (Basic) Counting Principle.	Grade 7, Mission 8, Topic B, Lesson 8-9 <i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
Patterns, Functions, and Algebra		

VIRGINIA'S MATHEMATICS STANDARDS OF LEARNING		ZEARN MATH
5th Grade Standards		Lessons
5.PFA.1a	Identify, describe, extend, and create increasing and decreasing patterns using various representations (e.g., objects, pictures, numbers, number lines, input/output tables, function machines).	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
5.PFA.1b	Analyze an increasing or decreasing single-operation numerical pattern found in lists, input/output tables, and function machines, and generalize the change to identify the rule, extend the pattern, or identify missing terms. (Patterns will be limited to addition, subtraction, multiplication, and division of whole numbers; addition and subtraction of fractions with like denominators of 12 or less; and addition and subtraction of decimals expressed in tenths or hundredths).	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
5.PFA.1c	Solve contextual problems that involve identifying, describing, and extending increasing and decreasing patterns using single-operation input and output rules (limited to addition, subtraction, multiplication, and division of whole numbers; addition and subtraction of fractions with like denominators of 12 or less; and addition and subtraction of decimals expressed in tenths or hundredths).	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
5.PFA.2a	Describe the concept of a variable (presented as a box, letter, or other symbol) as a representation of an unknown quantity.	Grade 6, Mission 6, Topic A, Lesson 2
5.PFA.2b	Write an equation (with a single variable that represents an unknown quantity and one operation) from a contextual situation, using addition, subtraction, multiplication, or division.	Grade 6, Mission 6, Topic A, Lesson 3-5 Grade 6, Mission 6, Topic B, Lesson 7
5.PFA.2c	Use an expression with a variable to represent a given verbal expression involving one operation (e.g., "5 more than a number" can be represented by $y + 5$).	Grade 6, Mission 1, Topic B, Lesson 5 Grade 6, Mission 1, Topic C, Lesson 9 Grade 6, Mission 1, Topic F, Lesson 18 Grade 6, Mission 6, Topic B, Lesson 6

VIRGINIA'S MATHEMATICS STANDARDS OF LEARNING		ZEARN MATH
5th Grade Standards		Lessons
5.PFA.2d	Create and write a word problem to match a given equation with a single variable and one operation.	Grade 6, Mission 6, Topic A, Lesson 3-5 Grade 6, Mission 6, Topic B, Lesson 7

6th Grade

VIRGINIA'S MATHEMATICS STANDARDS OF LEARNING		ZEARN MATH
6th Grade Standards		Lessons
Number and Number Sense		
6.NS.1a	Estimate and determine the percent represented by a given model (e.g., number line, picture, verbal description), including percents greater than 100% and less than 1%.	Mission 3, Topic D, Lesson 10-16 Mission 6, Topic D, Lesson 7
6.NS.1b	Represent and determine equivalencies among decimals (through the thousandths place) and percents incorporating the use of number lines, and concrete and pictorial models.	Mission 3, Topic D, Lesson 10, 14-16
6.NS.1c	Represent and determine equivalencies among fractions (proper or improper) and mixed numbers that have denominators that are 12 or less or factors of 100 and percents incorporating the use of number lines, and concrete and pictorial models.	Mission 3, Topic D, Lesson 11, 13-16
6.NS.1d	Represent and determine equivalencies among decimals, percents, fractions (proper or improper), and mixed numbers that have denominators that are 12 or less or factors of 100 incorporating the use of number lines, and concrete and pictorial models.	Mission 3, Topic D, Lesson 13-16
6.NS.1e	Use multiple strategies (e.g., benchmarks, number line, equivalency) to compare and order no more than four positive rational numbers expressed as fractions (proper or improper), mixed numbers, decimals, and percents (decimals through thousandths, fractions with denominators of 12 or less or factors of 100) with and without models. Justify solutions orally, in writing or with a model. Ordering may be in ascending or descending order.	Mission 7, Topic A, Lesson 4
6.NS.2a	Represent integers (e.g., number lines, concrete materials, pictorial models), including models derived from contextual situations, and identify an integer represented by a point on a number line.	Mission 7, Topic A, Lesson 1-2, 4-5, 7 Mission 7, Topic C, Lesson 11-15

VIRGINIA'S MATHEMATICS STANDARDS OF LEARNING		ZEARN MATH
6th Grade Standards		Lessons
6.NS.2b	Compare and order integers using a number line.	Mission 7, Topic A, Lesson 3-4, 6-7 Mission 7, Topic B, Lesson 8-9
6.NS.2c	Compare integers, using mathematical symbols ($<$, $>$, $=$).	Mission 7, Topic A, Lesson 3-4, 6-7 Mission 7, Topic B, Lesson 8-9
6.NS.2d	Identify and describe the absolute value of an integer as the distance from zero on the number line.	Mission 7, Topic A, Lesson 6-7
6.NS.3a	Recognize and represent patterns with bases and exponents that are whole numbers.	Mission 1, Topic F, Lesson 17 Mission 6, Topic C, Lesson 12-15
6.NS.3b	Recognize and represent patterns of perfect squares not to exceed 202, by using concrete and pictorial models.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
6.NS.3c	Justify if a number between 0 and 400 is a perfect square through modeling or mathematical reasoning.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
6.NS.3d	Recognize and represent powers of 10 with whole number exponents by examining patterns in place value.	Mission 1, Topic A, Lesson 3-4 Mission 2, Topic D, Lesson 13
Computation and Estimation		
6.CE.1a	Demonstrate/model multiplication and division of fractions (proper or improper) and mixed numbers using multiple representations.	Mission 4, Topic A, Lesson 3 Mission 4, Topic B, Lesson 4-9 Mission 4, Topic C, Lesson 10-11 Mission 4, Topic D, Lesson 12-14 Mission 4, Topic E, Lesson 16-17
6.CE.1b	Multiply and divide fractions (proper or improper) and mixed numbers that include denominators of 12 or less. Answers are expressed in simplest form.	Mission 4, Topic A, Lesson 3 Mission 4, Topic B, Lesson 4-9 Mission 4, Topic C, Lesson 10-11 Mission 4, Topic D, Lesson 12-14 Mission 4, Topic E, Lesson 16-17
6.CE.1c	Investigate and explain the effect of multiplying or dividing a fraction, whole number, or mixed number by a number between zero and one.	Grade 5, Mission 4, Topic F, Lesson 21-23

VIRGINIA'S MATHEMATICS STANDARDS OF LEARNING		ZEARN MATH
6th Grade Standards		Lessons
6.CE.1d	Estimate, determine, and justify the solution to single-step and multistep problems in context that involve addition and subtraction with fractions (proper or improper) and mixed numbers, with and without regrouping, that include like and unlike denominators of 12 or less. Answers are expressed in simplest form.	Grade 5, Mission 3, Topic D, Lesson 15 Grade 5, Mission 4, Topic D, Lesson 11-12
6.CE.1e	Estimate, determine, and justify the solution to single-step and multistep problems in context that involve multiplication and division with fractions (proper or improper) and mixed numbers that include denominators of 12 or less. Answers are expressed in simplest form.	Grade 5, Mission 4, Topic C, Lesson 6-7 Grade 5, Mission 4, Topic D, Lesson 10-12 Grade 5, Mission 4, Topic E, Lesson 13-18 Grade 5, Mission 4, Topic F, Lesson 22-24 Grade 5, Mission 5, Topic C, Lesson 12-15 Mission 4, Topic A, Lesson 3 Mission 4, Topic B, Lesson 4-9 Mission 4, Topic C, Lesson 10-11 Mission 4, Topic D, Lesson 12-14 Mission 4, Topic E, Lesson 16-17
6.CE.2a	Demonstrate/model addition, subtraction, multiplication, and division of integers using pictorial representations or concrete manipulatives.	Grade 7, Mission 5, Topic A, Lesson 2 Grade 7, Mission 5, Topic B, Lesson 5, 8
6.CE.2b	Add, subtract, multiply, and divide two integers.	Grade 7, Mission 5, Topic A, Lesson 2 Grade 7, Mission 5, Topic B, Lesson 5, 8
6.CE.2c	Simplify an expression that contains absolute value bars $ $ and an operation with two integers (e.g., $- 5 - 8 $ or $ -12 /8$) and represent the result on a number line.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
6.CE.2d	Estimate, determine, and justify the solution to one and two-step contextual problems, involving addition, subtraction, multiplication, and division with integers.	Grade 7, Mission 5, Topic A, Lesson 2 Grade 7, Mission 5, Topic B, Lesson 5, 8

VIRGINIA'S MATHEMATICS STANDARDS OF LEARNING		ZEARN MATH
6th Grade Standards		Lessons
Measurement and Geometry		
6.MG.1a	Identify and describe chord, diameter, radius, circumference, and area of a circle.	Grade 7, Mission 3, Topic A, Lesson 2
6.MG.1b	Investigate and describe the relationship between: i. diameter and radius; ii. radius and circumference; and iii. diameter and circumference.	Grade 7, Mission 3, Topic A, Lesson 2-5
6.MG.1c	Develop an approximation for pi (3.14) by gathering data and comparing the circumference to the diameter of various circles, using concrete manipulatives or technological models.	Grade 7, Mission 3, Topic A, Lesson 3-5
6.MG.1d	Develop the formula for circumference using the relationship between diameter, radius, and pi.	Grade 7, Mission 3, Topic A, Lesson 3-5
6.MG.1e	Solve problems, including those in context, involving circumference and area of a circle when given the length of the diameter or radius.	Grade 7, Mission 3, Topic A, Lesson 4-5 Grade 7, Mission 3, Topic B, Lesson 6-9 Grade 7, Mission 3, Topic C, Lesson 10-11 Grade 7, Mission 9, Topic A, Lesson 4 Grade 7, Mission 9, Topic C, Lesson 11-12
6.MG.2a	Develop the formula for determining the area of parallelograms and triangles using pictorial representations and concrete manipulatives (e.g., two-dimensional diagrams, grid paper).	Mission 1, Topic B, Lesson 4-6 Mission 1, Topic C, Lesson 7-10
6.MG.2b	Solve problems, including those in context, involving the perimeter and area of triangles and parallelograms.	Mission 1, Topic A, Lesson 2-3 Mission 1, Topic B, Lesson 4-6 Mission 1, Topic C, Lesson 7-10
6.MG.3a	Identify and label the axes, origin, and quadrants of a coordinate plane.	Grade 5, Mission 6, Topic A, Lesson 1-4 Grade 5, Mission 6, Topic B, Lesson 8-9

VIRGINIA'S MATHEMATICS STANDARDS OF LEARNING		ZEARN MATH
6th Grade Standards		Lessons
6.MG.3b	Identify and describe the location (quadrant or the axis) of a point given as an ordered pair. Ordered pairs will be limited to coordinates expressed as integers.	Mission 7, Topic C, Lesson 11-14
6.MG.3c	Graph ordered pairs in the four quadrants and on the axes of a coordinate plane. Ordered pairs will be limited to coordinates expressed as integers.	Mission 7, Topic A, Lesson 2 Mission 7, Topic C, Lesson 11-15
6.MG.3d	Identify ordered pairs represented by points in the four quadrants and on the axes of the coordinate plane. Ordered pairs will be limited to coordinates expressed as integers.	Mission 7, Topic A, Lesson 2 Mission 7, Topic C, Lesson 11-15
6.MG.3e	Relate the coordinates of a point to the distance from each axis and relate the coordinates of a single point to another point on the same horizontal or vertical line. Ordered pairs will be limited to coordinates expressed as integers.	Mission 7, Topic C, Lesson 11, 14
6.MG.3f	Draw polygons in the coordinate plane given coordinates for the vertices; use coordinates to determine the length of a side joining points with the same first coordinate or the same second coordinate. Ordered pairs will be limited to coordinates expressed as integers. Apply these techniques in the context of solving contextual and mathematical problems.	Mission 7, Topic C, Lesson 15
6.MG.4a	Identify regular polygons.	Mission 1, Topic D, Lesson 11
6.MG.4b	Draw lines of symmetry to divide regular polygons into two congruent parts.	Grade 4, Mission 4, Topic D, Lesson 12
6.MG.4c	Determine the congruence of segments, angles, and polygons given their properties.	Grade 8, Mission 1, Topic C, Lesson 11-13 Grade 8, Mission 1, Topic D, Lesson 15 Grade 8, Mission 2, Topic B, Lesson 6-7
6.MG.4d	Determine whether polygons are congruent or non congruent according to the measures of their sides and angles.	Grade 8, Mission 1, Topic C, Lesson 11-13 Grade 8, Mission 1, Topic D, Lesson 15

VIRGINIA'S MATHEMATICS STANDARDS OF LEARNING		ZEARN MATH
6th Grade Standards		Lessons
		Grade 8, Mission 2, Topic B, Lesson 6-7
Probability and Statistics		
6.PS.1a	Formulate questions that require the collection or acquisition of data with a focus on circle graphs	Mission 8, Topic A, Lesson 1-3
6.PS.1b	Determine the data needed to answer a formulated question and collect the data (or acquire existing data) using various methods (e.g., observations, measurement, surveys, experiments).	Mission 8, Topic A, Lesson 1-3
6.PS.1c	Determine the factors that will ensure that the data collected is a sample that is representative of a larger population	Grade 7, Mission 8, Topic C, Lesson 12-14 Grade 7, Mission 8, Topic E, Lesson 20
6.PS.1d	Organize and represent data using circle graphs, with and without the use of technology tools. The number of data values should be limited to allow for comparisons that have denominators of 12 or less or those that are factors of 100 (e.g., in a class of 20 students, 7 choose apples as a favorite fruit, so the comparison is 7 out of 20, $\frac{7}{20}$, or 35%).	Mission 8, Topic B, Lesson 3
6.PS.1e	Analyze data represented in a circle graph by making observations and drawing conclusions.	Mission 8, Topic B, Lesson 3
6.PS.1f	Compare data represented in a circle graph with the same data represented in other graphs, including but not limited to bar graphs, pictographs, and line plots (dot plots), and justify which graphical representation best represents the data.	Mission 8, Topic B, Lesson 3
6.PS.2a	Represent the mean of a set of data graphically as the balance point represented in a line plot (dot plot).	Mission 8, Topic C, Lesson 9-10
6.PS.2b	Determine the effect on measures of center when a single value of a data set is added, removed, or changed.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>

VIRGINIA'S MATHEMATICS STANDARDS OF LEARNING		ZEARN MATH
6th Grade Standards		Lessons
6.PS.2c	Observe patterns in data to identify outliers and determine their effect on mean, median, mode, or range.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
Patterns, Functions, and Algebra		
6.PFA.1a	Represent a relationship between two quantities using ratios.	Mission 2, Topic A, Lesson 1-2 Mission 2, Topic B, Lesson 3-5 Mission 9, Topic A, Lesson 4
6.PFA.1b	Represent a relationship in context that makes a comparison by using the notations a/b , $a:b$, and a to b .	Mission 2, Topic A, Lesson 1-2 Mission 2, Topic B, Lesson 3-5 Mission 9, Topic A, Lesson 4
6.PFA.1c	Represent different comparisons within the same quantity or between different quantities (e.g., part to part, part to whole, whole to whole).	Mission 2, Topic E, Lesson 15 Mission 9, Topic A, Lesson 4
6.PFA.1d	Create a relationship in words for a given ratio expressed symbolically	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
6.PFA.1e	Create a table of equivalent ratios to represent a proportional relationship between two quantities, when given a ratio.	Mission 2, Topic D, Lesson 11-13 Mission 2, Topic E, Lesson 16 Mission 3, Topic C, Lesson 5-9 Grade 7, Mission 2, Topic A, Lesson 2-3 Grade 7, Mission 4, Topic A, Lesson 3 Grade 7, Mission 9, Topic A, Lesson 3 Grade 7, Mission 9, Topic B, Lesson 5
6.PFA.1f	Create a table of equivalent ratios to represent a proportional relationship between two quantities, when given a contextual situation.	Mission 2, Topic D, Lesson 11-13 Mission 2, Topic E, Lesson 16 Mission 3, Topic C, Lesson 5-9 Grade 7, Mission 2, Topic A, Lesson 2-3 Grade 7, Mission 4, Topic A, Lesson 3 Grade 7, Mission 9, Topic A, Lesson 3

VIRGINIA'S MATHEMATICS STANDARDS OF LEARNING		ZEARN MATH
6th Grade Standards		Lessons
		Grade 7, Mission 9, Topic B, Lesson 5
6.PFA.2a	Identify the unit rate of a proportional relationship represented by a table of values, a contextual situation, or a graph.	Mission 2, Topic C, Lesson 10 Mission 3, Topic A, Lesson 1 Mission 3, Topic C, Lesson 5-7, 9 Mission 9, Topic A, Lesson 6
6.PFA.2b	Determine a missing value in a ratio table that represents a proportional relationship between two quantities using a unit rate.	Mission 2, Topic D, Lesson 11-13
6.PFA.2c	Determine whether a proportional relationship exists between two quantities, when given a table of values, context, or graph.	Grade 7, Mission 2, Topic A, Lesson 2-3 Grade 7, Mission 2, Topic C, Lesson 10 Grade 7, Mission 3, Topic A, Lesson 2-3
6.PFA.2d	When given a contextual situation representing a proportional relationship, find the unit rate and create a table of values or a graph.	Grade 7, Mission 2, Topic A, Lesson 2-3 Grade 7, Mission 2, Topic B, Lesson 4 Grade 7, Mission 2, Topic D, Lesson 11
6.PFA.2e	Make connections between and among multiple representations of the same proportional relationship using verbal descriptions, ratio tables, and graphs.	Grade 7, Mission 2, Topic B, Lesson 5 Grade 7, Mission 2, Topic D, Lesson 13 Grade 7, Mission 2, Topic E, Lesson 14
6.PFA.3a	Identify and develop examples of the following algebraic vocabulary: equation, variable, expression, term, and coefficient.	Mission 6, Topic A, Lesson 2
6.PFA.3b	Represent and solve one-step linear equations in one variable, using a variety of concrete manipulatives and pictorial representations (e.g., colored chips, algebra tiles, weights on a balance scale).	Mission 6, Topic A, Lesson 3-5

VIRGINIA'S MATHEMATICS STANDARDS OF LEARNING		ZEARN MATH
6th Grade Standards		Lessons
6.PFA.3c	Apply properties of real numbers and properties of equality to solve a one-step equation in one variable. Coefficients are limited to integers and unit fractions. Numeric terms are limited to integers.	Mission 6, Topic A, Lesson 3-5
6.PFA.3d	Confirm solutions to one-step linear equations in one variable using a variety of concrete manipulatives and pictorial representations (e.g., colored chips, algebra tiles, weights on a balance scale).	Mission 6, Topic A, Lesson 2-5
6.PFA.3e	Write a one-step linear equation in one variable to represent a verbal situation, including those in context	Mission 6, Topic A, Lesson 4
6.PFA.3f	Create a verbal situation in context given a one-step linear equation in one variable.	Mission 6, Topic A, Lesson 4
6.PFA.4a	Given the graph of a linear inequality in one variable on a number line, represent the inequality in two equivalent ways (e.g., $x < -5$ or $-5 > x$) using symbols. Symbols include $<$, $>$, $\leq$, $\geq$.	Mission 7, Topic B, Lesson 8-10
6.PFA.4b	Write a linear inequality in one variable to represent a given constraint or condition in context or given a graph on a number line	Mission 7, Topic B, Lesson 8-10
6.PFA.4c	Given a linear inequality in one variable, create a corresponding contextual situation or create a number line graph.	Mission 7, Topic B, Lesson 10
6.PFA.4d	Use substitution or a number line graph to justify whether a given number in a specified set makes a linear inequality in one variable true	Mission 7, Topic B, Lesson 8-10
6.PFA.4e	Identify a numerical value(s) that is part of the solution set of a given inequality in one variable.	Mission 7, Topic B, Lesson 8-10

7th Grade

VIRGINIA'S MATHEMATICS STANDARDS OF LEARNING		ZEARN MATH
7th Grade Standards		Lessons
Number and Number Sense		
7.NS.1a	Investigate and describe powers of 10 with negative exponents by examining patterns.	Grade 8, Mission 7, Topic C, Lesson 9-12, 14 Grade 8, Mission 7, Topic D, Lesson 16
7.NS.1b	Represent a power of 10 with a negative exponent in fraction and decimal form.	Grade 8, Mission 7, Topic C, Lesson 9-12, 14 Grade 8, Mission 7, Topic D, Lesson 16
7.NS.1c	Convert between numbers greater than 0 written in scientific notation and decimals.	Grade 8, Mission 7, Topic C, Lesson 9-12, 14 Grade 8, Mission 7, Topic D, Lesson 16
7.NS.1d	Compare and order no more than four numbers greater than 0 written in scientific notation. Ordering may be in ascending or descending order.	Grade 8, Mission 7, Topic C, Lesson 11-12, 14 Grade 8, Mission 7, Topic D, Lesson 16
7.NS.2a	Use multiple strategies (e.g., benchmarks, number line, equivalency) to compare (using symbols $<$, $>$, $=$) and order (a set of no more than four) rational numbers expressed as integers, fractions (proper or improper), mixed numbers, decimals, and percents. Fractions and mixed numbers may be positive or negative. Decimals may be positive or negative and are limited to the thousandths place. Ordering may be in ascending or descending order. Justify solutions orally, in writing or with a model.	Grade 6, Mission 7, Topic A, Lesson 3 Grade 6, Mission 7, Topic B, Lesson 8
7.NS.3a	Determine the positive square root of a perfect square from 0 to 400.	Grade 8, Mission 8, Topic B, Lesson 2-5 Grade 8, Mission 8, Topic C, Lesson 10 Grade 8, Mission 8, Topic B, Lesson 12-13

VIRGINIA'S MATHEMATICS STANDARDS OF LEARNING		ZEARN MATH
7th Grade Standards		Lessons
7.NS.3b	Describe the relationship between square roots and perfect squares.	Grade 8, Mission 8, Topic B, Lesson 2-5 Grade 8, Mission 8, Topic C, Lesson 10 Grade 8, Mission 8, Topic B, Lesson 12-13
Computation and Estimation		
7.CE.1a	Estimate, solve, and justify solutions to contextual problems involving addition, subtraction, multiplication, and division with rational numbers expressed as integers, fractions (proper or improper), mixed numbers, and decimals. Fractions may be positive or negative. Decimals may be positive or negative and are limited to the thousandths place.	Mission 5, Topic D, Lesson 13-14 Mission 5, Topic E, Lesson 15-16 Mission 5, Topic F, Lesson 17 Mission 9, Topic A, Lesson 3 Mission 9, Topic B, Lesson 6
7.CE.2a	Given a proportional relationship between two quantities, create and use a ratio table to determine missing values.	Mission 2, Topic A, Lesson 2-3 Mission 2, Topic D, Lesson 10 Mission 3, Topic A, Lesson 1, 3, 5 Mission 3, Topic B, Lesson 7
7.CE.2b	Write and solve a proportion that represents a proportional relationship between two quantities to find a missing value, including problems in context.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
7.CE.2c	Apply proportional reasoning to solve problems in context, including converting units of measurement, when given the conversion factor.	Grade 6, Mission 2, Topic C, Lesson 6-10 Grade 6, Mission 2, Topic D, Lesson 11-14 Grade 6, Mission 2, Topic E, Lesson 15-16 Grade 6, Mission 2, Topic F, Lesson 17 Grade 6, Mission 3, Topic B, Lesson 3-4 Grade 6, Mission 3, Topic C, Lesson 5-9 Grade 6, Mission 6, Topic D, Lesson 16 Grade 6, Mission 6, Topic E, Lesson 17 Grade 6, Mission 9, Topic A, Lesson 4-6

VIRGINIA'S MATHEMATICS STANDARDS OF LEARNING		ZEARN MATH
7th Grade Standards		Lessons
7.CE.2d	Estimate and determine the percentage of a given whole number, including but not limited to the use of benchmark percentages.	Grade 6, Mission 3, Topic D, Lesson 10-15 Grade 6, Mission 9, Topic A, Lesson 4-6
Measurement and Geometry		
7.MG.1a	Develop the formulas for determining the volume of right cylinders and solve problems, including those in contextual situations, using concrete objects, diagrams, and formulas.	Grade 8, Mission 5, Topic D, Lesson 13-16 Grade 8, Mission 5, Topic E, Lesson 17-21 Grade 8, Mission 5, Topic F, Lesson 22
7.MG.1b	Develop the formulas for determining the surface area of rectangular prisms and right cylinders and solve problems, including those in contextual situations, using concrete objects, two-dimensional diagrams, nets, and formulas.	Mission 1, Topic A, Lesson 6 Mission 2, Topic C, Lesson 8 Mission 3, Topic B, Lesson 6 Mission 7, Topic C, Lesson 12-16 Mission 7, Topic D, Lesson 17 Mission 9, Topic A, Lesson 4 Mission 9, Topic B, Lesson 5, 9
7.MG.1c	Determine if a problem in context, involving a rectangular prism or right cylinder, represents the application of volume or surface area.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
7.MG.1d	Describe how the volume of a rectangular prism is affected when one measured attribute is multiplied by a factor of $\frac{1}{4}$, $\frac{1}{3}$, $\frac{1}{2}$, 2, 3, or 4, including those in contextual situations.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
7.MG.1e	Describe how the surface area of a rectangular prism is affected when one measured attribute is multiplied by a factor of $\frac{1}{2}$ or 2, including those in contextual situations.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
7.MG.2a	Identify corresponding congruent angles of similar quadrilaterals and triangles, through the use of geometric markings.	Mission 1, Topic A, Lesson 1-6 Mission 1, Topic B, Lesson 7, 9, 11
7.MG.2b	Identify corresponding sides of similar quadrilaterals and triangles.	Mission 1, Topic A, Lesson 1-6 Mission 1, Topic B, Lesson 7, 9, 11

VIRGINIA'S MATHEMATICS STANDARDS OF LEARNING		ZEARN MATH
7th Grade Standards		Lessons
7.MG.2c	Given two similar quadrilaterals or triangles, write similarity statements using symbols	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
7.MG.2d	Write proportions to express the relationships between the lengths of corresponding sides of similar quadrilaterals and triangles.	Grade 8, Mission 1, Topic C, Lesson 11-13 Grade 8, Mission 1, Topic D, Lesson 15 Grade 8, Mission 2, Topic B, Lesson 6-7
7.MG.2e	Recognize and justify if two quadrilaterals or triangles are similar using the ratios of corresponding side lengths.	Grade 8, Mission 1, Topic C, Lesson 11-13 Grade 8, Mission 1, Topic D, Lesson 15 Grade 8, Mission 2, Topic B, Lesson 6-7
7.MG.2f	Solve a proportion to determine a missing side length of similar quadrilaterals or triangles.	Grade 8, Mission 2, Topic B, Lesson 6-7, 9
7.MG.2g	Given angle measures in a quadrilateral or triangle, determine unknown angle measures in a similar quadrilateral or triangle.	Grade 8, Mission 2, Topic B, Lesson 6-7, 9
7.MG.2h	Apply proportional reasoning to solve problems in context including scale drawings. Scale factors shall have denominators no greater than 12 and decimals no less than tenths.	Mission 1, Topic A, Lesson 1-6 Mission 1, Topic B, Lesson 7-12 Mission 1, Topic C, Lesson 13 Mission 2, Topic A, Lesson 1 Mission 3, Topic B, Lesson 6 Mission 3, Topic C, Lesson 11 Mission 9, Topic A, Lesson 4 Mission 9, Topic C, Lesson 13
7.MG.3a	Compare and contrast properties of the following quadrilaterals: parallelogram, rectangle, square, rhombus, and trapezoid: i. parallel/perpendicular sides and diagonals; ii. congruence of angle measures, side, and diagonal lengths; and iii. lines of symmetry.	Grade 5, Mission 5, Topic D, Lesson 18-21

VIRGINIA'S MATHEMATICS STANDARDS OF LEARNING		ZEARN MATH
7th Grade Standards		Lessons
7.MG.3b	Sort and classify quadrilaterals as parallelograms, rectangles, trapezoids, rhombi, and/or squares based on their properties: i. parallel/perpendicular sides and diagonals; ii. congruence of angle measures, side, and diagonal lengths; and iii. lines of symmetry.	Grade 5, Mission 5, Topic C, Lesson 16-17 Grade 5, Mission 5, Topic D, Lesson 18-21
7.MG.3c	Given a diagram, determine an unknown angle measure in a quadrilateral, using properties of quadrilaterals.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
7.MG.3d	Given a diagram, determine an unknown side length in a quadrilateral using properties of quadrilaterals.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
7.MG.4a	Given a preimage in the coordinate plane, identify the coordinates of the image of a polygon that has been dilated. Scale factors are limited to $\frac{1}{4}$, $\frac{1}{2}$, 2, 3, or 4. The center of the dilation will be the origin	Grade 8, Mission 1, Topic A, Lesson 5-6 Grade 8, Mission 2, Topic A, Lesson 4-5 Grade 8, Mission 2, Topic C, Lesson 12
7.MG.4b	Sketch the image of a dilation of a polygon limited to a scale factor of $\frac{1}{4}$, $\frac{1}{2}$, 2, 3, or 4. The center of the dilation will be the origin.	Grade 8, Mission 1, Topic A, Lesson 5-6 Grade 8, Mission 2, Topic A, Lesson 4-5 Grade 8, Mission 2, Topic C, Lesson 12
7.MG.4c	Identify and describe dilations in context including, but not limited to, scale drawings and graphic design.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
Probability and Statistics		
7.PS.1a	Determine the theoretical probability of an event.	Mission 8, Topic A, Lesson 1, 3-6
7.PS.1b	Given the results of a statistical investigation, determine the experimental probability of an event.	Mission 8, Topic A, Lesson 1, 3-6
7.PS.1c	Describe changes in the experimental probability as the number of trials increases.	Mission 8, Topic A, Lesson 1, 3-6

VIRGINIA'S MATHEMATICS STANDARDS OF LEARNING		ZEARN MATH
7th Grade Standards		Lessons
7.PS.1d	Investigate and describe the difference between the probability of an event found through experiment or simulation versus the theoretical probability of that same event.	Mission 8, Topic A, Lesson 1, 3-6
7.PS.2a	Formulate questions that require the collection or acquisition of data with a focus on histograms.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
7.PS.2b	Determine the data needed to answer a formulated question and collect the data (or acquire existing data) using various methods (e.g., observations, measurement, surveys, experiments).	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
7.PS.2c	Determine how sample size and randomness will ensure that the data collected is a sample that is representative of a larger population.	Mission 8, Topic C, Lesson 12-14 Mission 8, Topic D, Lesson 15 Mission 8, Topic E, Lesson 20
7.PS.2d	Organize and represent numerical data using histograms with and without the use of technology.	Grade 6, Mission 8, Topic B, Lesson 3-8 Grade 6, Mission 8, Topic D, Lesson 16-17
7.PS.2e	Investigate and explain how using different intervals could impact the representation of the data in a histogram.	Grade 6, Mission 8, Topic B, Lesson 3-8
7.PS.2f	Compare data represented in histograms with the same data represented in other graphs, including but not limited to line plots (dot plots), circle graphs, and stem-and-leaf plots, and justify which graphical representation best represents the data.	Grade 6, Mission 8, Topic B, Lesson 3-8
7.PS.2g	Analyze data represented in histograms by making observations and drawing conclusions. Determine how histograms reveal patterns in data that cannot be easily seen by looking at the corresponding given data set.	Grade 6, Mission 8, Topic B, Lesson 3-8
Patterns, Functions, and Algebra		
7.PFA.1a	Determine the slope, m , as the rate of change in a proportional relationship between two quantities	Grade 8, Mission 3, Topic A, Lesson 2-4

VIRGINIA'S MATHEMATICS STANDARDS OF LEARNING		ZEARN MATH
7th Grade Standards		Lessons
	given a table of values, graph, or contextual situation and write an equation in the form $y = mx$ to represent the direct variation relationship. Slope may include positive or negative values (slope will be limited to positive values in a contextual situation).	Grade 8, Mission 3, Topic B, Lesson 6
7.PFA.1b	Identify and describe a line with a slope that is positive, negative, or zero (0), given a graph.	Grade 8, Mission 2, Topic C, Lesson 10-12 Grade 8, Mission 3, Topic B, Lesson 7 Grade 8, Mission 3, Topic C, Lesson 10-11 Grade 8, Mission 3, Topic E, Lesson 14
7.PFA.1c	Graph a line representing a proportional relationship, between two quantities given an ordered pair on the line and the slope, m , as rate of change. Slope may include positive or negative values	Grade 8, Mission 3, Topic A, Lesson 2-4 Grade 8, Mission 3, Topic B, Lesson 6
7.PFA.1d	Graph a line representing a proportional relationship between two quantities given the equation of the line in the form $y = mx$, where m represents the slope as rate of change. Slope may include positive or negative values.	Grade 8, Mission 3, Topic A, Lesson 2-4 Grade 8, Mission 3, Topic B, Lesson 6
7.PFA.1e	Make connections between and among representations of a proportional relationship between two quantities using problems in context, tables, equations, and graphs. Slope may include positive or negative values (slope will be limited to positive values in a contextual situation).	Grade 8, Mission 3, Topic A, Lesson 2-4 Grade 8, Mission 3, Topic B, Lesson 6
7.PFA.2a	Use the order of operations and apply the properties of real numbers to simplify numerical expressions. Exponents are limited to 1, 2, 3, or 4 and bases are limited to positive integers. Expressions should not include braces $\{ \}$ but may include brackets $[]$ and absolute value bars $ $. Square roots are limited to perfect squares.	Grade 6, Mission 1, Topic F, Lesson 17-18 Grade 6, Mission 6, Topic C, Lesson 12-15
7.PFA.2b	Represent equivalent algebraic expressions in one variable using concrete manipulatives and pictorial representations (e.g., colored chips, algebra tiles).	Grade 6, Mission 1, Topic B, Lesson 5 Grade 6, Mission 1, Topic C,

VIRGINIA'S MATHEMATICS STANDARDS OF LEARNING		ZEARN MATH
7th Grade Standards		Lessons
		Lesson 9 Grade 6, Mission 1, Topic F, Lesson 18 Grade 6, Mission 6, Topic B, Lesson 6
7.PFA.2c	Simplify and generate equivalent algebraic expressions in one variable by applying the order of operations and properties of real numbers. Expressions may require combining like terms to simplify. Expressions will include only linear and numeric terms. Coefficients and numeric terms may be positive or negative rational numbers.	Grade 6, Mission 6, Topic B, Lesson 10-11
7.PFA.2d	Use the order of operations and apply the properties of real numbers to evaluate algebraic expressions for given replacement values of the variables. Exponents are limited to 1, 2, 3, or 4 and bases are limited to positive integers. Expressions should not include braces { } but may include brackets [] and absolute value bars . Square roots are limited to perfect squares. Limit the number of replacements to no more than three per expression. Replacement values may be positive or negative rational numbers.	Grade 6, Mission 1, Topic B, Lesson 5-6 Grade 6, Mission 1, Topic C, Lesson 9-10 Grade 6, Mission 6, Topic B, Lesson 6 Grade 6, Mission 6, Topic C, Lesson 14-15
7.PFA.3a	Represent and solve two-step linear equations in one variable using a variety of concrete materials and pictorial representations.	Mission 5, Topic E, Lesson 15-16 Mission 6, Topic A, Lesson 4-5 Mission 6, Topic B, Lesson 7-12 Mission 9, Topic B, Lesson 7
7.PFA.3b	Apply properties of real numbers and properties of equality to solve two-step linear equations in one variable. Coefficients and numeric terms will be rational.	Mission 5, Topic E, Lesson 15-16 Mission 6, Topic A, Lesson 4-5 Mission 6, Topic B, Lesson 7-12 Mission 9, Topic B, Lesson 7
7.PFA.3c	Confirm algebraic solutions to linear equations in one variable.	Mission 5, Topic E, Lesson 15-16 Mission 6, Topic A, Lesson 4-5 Mission 6, Topic B, Lesson 7-12 Mission 9, Topic B, Lesson 7
7.PFA.3d	Write a two-step linear equation in one variable to represent a verbal situation, including those in context.	Mission 5, Topic E, Lesson 15-16 Mission 6, Topic A, Lesson 4-5 Mission 6, Topic B, Lesson 7-12 Mission 9, Topic B, Lesson 7

VIRGINIA'S MATHEMATICS STANDARDS OF LEARNING		ZEARN MATH
7th Grade Standards		Lessons
7.PFA.3e	Create a verbal situation in context given a two-step linear equation in one variable.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
7.PFA.3f	Solve problems in context that require the solution of a two-step linear equation	Mission 5, Topic E, Lesson 15-16 Mission 6, Topic A, Lesson 4-5 Mission 6, Topic B, Lesson 7-12 Mission 9, Topic B, Lesson 7
7.PFA.4a	Apply properties of real numbers and the addition, subtraction, multiplication, and division properties of inequality to solve one- and two-step inequalities in one variable. Coefficients and numeric terms will be rational.	Mission 6, Topic C, Lesson 14, 16-17
7.PFA.4b	Investigate and explain how the solution set of a linear inequality is affected by multiplying or dividing both sides of the inequality statement by a rational number less than zero	Mission 6, Topic C, Lesson 14, 16-17
7.PFA.4c	Represent solutions to one- or two-step linear inequalities in one variable algebraically and graphically using a number line.	Mission 6, Topic C, Lesson 14, 16-17
7.PFA.4d	Write one- or two-step linear inequalities in one variable to represent a verbal situation, including those in context.	Mission 6, Topic C, Lesson 14, 16-17
7.PFA.4e	Create a verbal situation in context given a one or two-step linear inequality in one variable.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
7.PFA.4f	Solve problems in context that require the solution of a one- or two-step inequality.	Mission 6, Topic C, Lesson 14, 16-17
7.PFA.4g	Identify a numerical value(s) that is part of the solution set of as given one- or two-step linear inequality in one variable.	Mission 6, Topic C, Lesson 14, 16-17
7.PFA.4h	Describe the differences and similarities between solving linear inequalities in one variable and linear equations in one variable.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>

8th Grade

VIRGINIA'S MATHEMATICS STANDARDS OF LEARNING		ZEARN MATH
8th Grade Standards		Lessons
Number and Number Sense		
8.NS.1a	Estimate and identify the two consecutive natural numbers between which the positive square root of a given number lies and justify which natural number is the better approximation. Numbers are limited to natural numbers from 1 to 400.	Mission 8, Topic B, Lesson 4-5 Mission 8, Topic E, Lesson 15
8.NS.1b	Use rational approximations (to the nearest hundredth) of irrational numbers to compare, order, and locate values on a number line. Radicals may include both positive and negative square roots of values from 0 to 400 yielding an irrational number.	Mission 8, Topic B, Lesson 4-5 Mission 8, Topic E, Lesson 15
8.NS.1c	Use multiple strategies (e.g., benchmarks, number line, equivalency) to compare and order no more than five real numbers expressed as integers, fractions (proper or improper), decimals, mixed numbers, percents, numbers written in scientific notation, radicals, and π . Radicals may include both positive and negative square roots of values from 0 to 400. Ordering may be in ascending or descending order. Justify solutions orally, in writing or with a model.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
8.NS.2a	Describe and illustrate the relationships among the subsets of the real number system by using representations (e.g., graphic organizers, number lines). Subsets include rational numbers, irrational numbers, integers, whole numbers, and natural numbers.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
8.NS.2b	Classify and explain why a given number is a member of a particular subset or subsets of the real number system.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
8.NS.2c	Describe each subset of the set of real numbers and include examples and non-examples.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>

VIRGINIA'S MATHEMATICS STANDARDS OF LEARNING		ZEARN MATH
8th Grade Standards		Lessons
Computation and Estimation		
8.CE.1a	Estimate and solve contextual problems that require the computation of one discount or markup and the resulting sale price.	Grade 7, Mission 4, Topic B, Lesson 6-7 Grade 7, Mission 4, Topic C, Lesson 11-12 Grade 7, Mission 9, Topic A, Lesson 3
8.CE.1b	Estimate and solve contextual problems that require the computation of the sales tax, tip and resulting total.	Grade 7, Mission 4, Topic C, Lesson 10-12
8.CE.1c	Estimate and solve contextual problems that require the computation of the percent increase or decrease.	Grade 7, Mission 4, Topic B, Lesson 6-9 Grade 7, Mission 4, Topic C, Lesson 10-15 Grade 7, Mission 4, Topic D, Lesson 16
Measurement and Geometry		
8.MG.1a	Identify and describe the relationship between pairs of angles that are vertical, adjacent, supplementary, and complementary.	Grade 7, Mission 7, Topic A, Lesson 1-5
8.MG.1b	Use the relationships among supplementary, complementary, vertical, and adjacent angles to solve problems, including those in context, involving the measure of unknown angles.	Grade 7, Mission 7, Topic A, Lesson 1-5
8.MG.2a	Determine the surface area of square-based pyramids by using concrete objects, nets, diagrams, and formulas.	Grade 6, Mission 1, Topic E, Lesson 14-15
8.MG.2b	Determine the volume of cones and square-based pyramids, using concrete objects, diagrams, and formulas.	Mission 5, Topic D, Lesson 15 Mission 5, Topic E, Lesson 21
8.MG.2c	Examine and explain the relationship between the volume of cones and cylinders, and the volume of rectangular prisms and square based pyramids.	Mission 5, Topic D, Lesson 12-16 Mission 5, Topic E, Lesson 21

VIRGINIA'S MATHEMATICS STANDARDS OF LEARNING		ZEARN MATH
8th Grade Standards		Lessons
8.MG.2d	Solve problems in context involving volume of cones and square-based pyramids and the surface area of square-based pyramids.	Mission 5, Topic D, Lesson 15
8.MG.3a	Given a preimage in the coordinate plane, identify the coordinates of the image of a polygon that has been translated vertically, horizontally, or a combination of both.	Mission 1, Topic A, Lesson 5-6
8.MG.3b	Given a preimage in the coordinate plane, identify the coordinates of the image of a polygon that has been reflected over the x - or y -axis.	Mission 1, Topic A, Lesson 5-6
8.MG.3c	Given a preimage in the coordinate plane, identify the coordinates of the image of a polygon that has been translated and reflected over the x - or y -axis or reflected over the x - or y -axis and then translated.	Mission 1, Topic A, Lesson 5-6
8.MG.3d	Sketch the image of a polygon that has been translated vertically, horizontally, or a combination of both.	Mission 1, Topic A, Lesson 2-4, 6
8.MG.3e	Sketch the image of a polygon that has been reflected over the x - or y -axis.	Mission 1, Topic A, Lesson 2-4, 6
8.MG.3f	Sketch the image of a polygon that has been translated and reflected over the x - or y -axis, or reflected over the x - or y -axis and then translated.	Mission 1, Topic A, Lesson 2-4, 6
8.MG.3g	Identify and describe transformations in context (e.g., tiling, fabric, wallpaper designs, art).	Mission 1, Topic E, Lesson 17
8.MG.4a	Verify the Pythagorean Theorem using diagrams, concrete materials, and measurement.	Mission 8, Topic C, Lesson 7
8.MG.4b	Determine whether a triangle is a right triangle given the measures of its three sides.	Mission 8, Topic C, Lesson 6-7
8.MG.4c	Identify the parts of a right triangle (the hypotenuse and the legs) given figures in various orientations.	Mission 8, Topic C, Lesson 6-8

VIRGINIA'S MATHEMATICS STANDARDS OF LEARNING		ZEARN MATH
8th Grade Standards		Lessons
8.MG.4d	Determine the measure of a side of a right triangle, given the measures of the other two sides.	Mission 8, Topic C, Lesson 7-8
8.MG.4e	Apply the Pythagorean Theorem, and its converse, to solve problems involving right triangles in context.	Mission 8, Topic C, Lesson 6-10
8.MG.5a	Subdivide a plane figure into triangles, rectangles, squares, trapezoids, parallelograms, circles, and semicircles. Determine the area of subdivisions and combine to determine the area of the composite plane figure.	Grade 7, Mission 3, Topic A, Lesson 4 Grade 7, Mission 3, Topic B, Lesson 6 Grade 7, Mission 9, Topic A, Lesson 4
8.MG.5b	Subdivide a plane figure into triangles, rectangles, squares, trapezoids, parallelograms, and semicircles. Use the attributes of the subdivisions to determine the perimeter of the composite plane figure.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
8.MG.5c	Apply perimeter, circumference, and area formulas to solve contextual problems involving composite plane figures.	Grade 7, Mission 3, Topic A, Lesson 4 Grade 7, Mission 3, Topic B, Lesson 6 Grade 7, Mission 3, Topic C, Lesson 11 Grade 7, Mission 9, Topic A, Lesson 4
Probability and Statistics		
8.PS.1a	Determine whether two events are independent or dependent and explain how replacement impacts the probability.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
8.PS.1b	Compare and contrast the probability of independent and dependent events.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
8.PS.1c	Determine the probability of two independent events.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>

VIRGINIA'S MATHEMATICS STANDARDS OF LEARNING		ZEARN MATH
8th Grade Standards		Lessons
8.PS.1d	Determine the probability of two dependent events.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
8.PS.2a	Formulate questions that require the collection or acquisition of data with a focus on boxplots	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
8.PS.2b	Determine the data needed to answer a formulated question and collect the data (or acquire existing data) using various methods (e.g., observations, measurement, surveys, experiments).	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
8.PS.2c	Determine how statistical bias might affect whether the data collected from the sample is representative of the larger population.	Grade 7, Mission 8, Topic C, Lesson 12-14 Grade 7, Mission 8, Topic D, Lesson 15 Grade 7, Mission 8, Topic E, Lesson 20
8.PS.2d	Organize and represent a numeric data set of no more than 20 items, using boxplots, with and without the use of technology	Grade 6, Mission 8, Topic D, Lesson 16-17
8.PS.2e	Identify and describe the lower extreme (minimum), upper extreme (maximum), median, upper quartile, lower quartile, range, and interquartile range given a data set, represented by a boxplot	Grade 6, Mission 8, Topic D, Lesson 16-17
8.PS.2f	Describe how the presence of an extreme data point (outlier) affects the shape and spread of the data distribution of a boxplot.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
8.PS.2g	Analyze data represented in a boxplot by making observations and drawing conclusions.	Grade 6, Mission 8, Topic D, Lesson 16-17 Grade 6, Mission 8, Topic E, Lesson 18
8.PS.2h	Compare and analyze two data sets represented in boxplots.	Grade 6, Mission 8, Topic D, Lesson 16-17 Grade 6, Mission 8, Topic E, Lesson 18

VIRGINIA'S MATHEMATICS STANDARDS OF LEARNING		ZEARN MATH
8th Grade Standards		Lessons
8.PS.2i	Given a contextual situation, justify which graphical representation (e.g., pictographs, bar graphs, line graphs, line plots/dot plots, stem-and-leaf plots, circle graphs, histograms, and boxplots) best represents the data.	Grade 6, Mission 8, Topic E, Lesson 18
8.PS.2j	Identify components of graphical displays that can be misleading.	Grade 6, Mission 8, Topic D, Lesson 14 Grade 6, Mission 8, Topic E, Lesson 18
8.PS.3a	Formulate questions that require the collection or acquisition of data with a focus on scatterplots.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
8.PS.3b	Determine the data needed to answer a formulated question and collect the data (or acquire existing data) of no more than 20 items using various methods (e.g., observations, measurement, surveys, experiments).	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
8.PS.3c	Organize and represent numeric bivariate data using scatterplots with and without the use of technology.	Mission 6, Topic A, Lesson 1-2 Mission 6, Topic B, Lesson 3-8
8.PS.3d	Make observations about a set of data points in a scatterplot as having a positive linear relationship, a negative linear relationship, or no relationship.	Mission 6, Topic B, Lesson 3-8
8.PS.3e	Analyze and justify the relationship of the quantitative bivariate data represented in scatterplots.	Mission 6, Topic B, Lesson 4-6, 8
8.PS.3f	Sketch the line of best fit for data represented in a scatterplot	Mission 6, Topic B, Lesson 5-6, 8
Patterns, Functions, and Algebra		
8.PFA.1a	Represent algebraic expressions using concrete manipulatives or pictorial representations (e.g., colored chips, algebra tiles), including expressions that apply the distributive property.	Grade 6, Mission 6, Topic B, Lesson 10-11
8.PFA.1b	Simplify and generate equivalent algebraic expressions in one variable by applying the order of	Grade 6, Mission 6, Topic B, Lesson 10-11

VIRGINIA'S MATHEMATICS STANDARDS OF LEARNING		ZEARN MATH
8th Grade Standards		Lessons
	operations and properties of real numbers. Expressions may need to be expanded (using the distributive property) or require combining like terms to simplify. Expressions will include only linear and numeric terms. Coefficients and numeric terms may be rational.	
8.PFA.2a	Determine whether a relation, represented by a set of ordered pairs, a table, or a graph of discrete points is a function. Sets are limited to no more than 10 ordered pairs.	Mission 5, Topic A, Lesson 1-2 Mission 5, Topic B, Lesson 3-5 Mission 5, Topic E, Lesson 17 Mission 9, Topic A, Lesson 1
8.PFA.2b	Identify the domain and range of a function represented as a set of ordered pairs, a table, or a graph of discrete points.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
8.PFA.3a	Determine how adding a constant (b) to the equation of a proportional relationship $y = mx$ will translate the line on a graph.	Mission 3, Topic B, Lesson 8
8.PFA.3b	Describe key characteristics of linear functions including slope (m), y -intercept (b), and independent and dependent variables.	Mission 5, Topic C, Lesson 8
8.PFA.3c	Graph a linear function given a table, equation, or a situation in context.	Mission 5, Topic C, Lesson 8-9
8.PFA.3d	Create a table of values for a linear function given a graph, equation in the form of $y = mx + b$, or context.	Mission 5, Topic C, Lesson 8
8.PFA.3e	Write an equation of a linear function in the form $y = mx + b$, given a graph, table, or a situation in context.	Mission 5, Topic C, Lesson 8-9
8.PFA.3f	Create a context for a linear function given a graph, table, or equation in the form $y = mx + b$.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
8.PFA.4a	Represent and solve multistep linear equations in one variable with the variable on one or both sides of the equation (up to four steps) using a variety of concrete materials and pictorial representations	Mission 4, Topic B, Lesson 3-9

VIRGINIA'S MATHEMATICS STANDARDS OF LEARNING		ZEARN MATH
8th Grade Standards		Lessons
8.PFA.4b	Apply properties of real numbers and properties of equality to solve multistep linear equations in one variable (up to four steps). Coefficients and numeric terms will be rational. Equations may contain expressions that need to be expanded (using the distributive property) or require combining like terms to solve.	Mission 4, Topic B, Lesson 3-9
8.PFA.4c	Write a multistep linear equation in one variable to represent a verbal situation, including those in context.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
8.PFA.4d	Create a verbal situation in context given a multistep linear equation in one variable.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
8.PFA.4e	Solve problems in context that require the solution of a multistep linear equation	Mission 4, Topic B, Lesson 9
8.PFA.4f	Interpret algebraic solutions in context to linear equations in one variable.	Mission 4, Topic B, Lesson 9
8.PFA.4g	Confirm algebraic solutions to linear equations in one variable.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
8.PFA.5a	Apply properties of real numbers and properties of inequality to solve multi step linear inequalities (up to four steps) in one variable with the variable on one or both sides of the inequality. Coefficients and numeric terms will be rational. Inequalities may contain expressions that need to be expanded (using the distributive property) or require combining like terms to solve.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
8.PFA.5b	Represent solutions to inequalities algebraically and graphically using a number line	Grade 7, Mission 6, Topic C, Lesson 14, 16-17 <i>It is recommended that teachers use supplemental materials to fully address this standard.</i>

VIRGINIA'S MATHEMATICS STANDARDS OF LEARNING		ZEARN MATH
8th Grade Standards		Lessons
8.PFA.5c	Write multi step linear inequalities in one variable to represent a verbal situation, including those in context.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
8.PFA.5d	Create a verbal situation in context given a multistep linear inequality in one variable	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
8.PFA.5e	Solve problems in context that require the solution of a multistep linear inequality in one variable.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
8.PFA.5f	Identify a numerical value(s) that is part of the solution set of a given inequality.	Grade 7, Mission 6, Topic C, Lesson 14, 16-17 <i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
8.PFA.5g	Interpret algebraic solutions in context to linear inequalities in one variable.	Grade 7, Mission 6, Topic C, Lesson 14, 16-17 <i>It is recommended that teachers use supplemental materials to fully address this standard.</i>